ESIA NON-TECHNICAL SUMMARY FOR TIN CAN AND APAPA PORTS MODERNIZATION PROJECT

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DOCUMENT DETAILS

Document Title:	Environmental and Social Impact Assessment (ESIA) Non-Techincal Summary for the Tin Can and Apapa Ports Modernization Project in Lagos, Nigeria
Document Subtitle	First Draft
Date:	February 2026
Version	01
Author	SkyKapital Europe
Client:	ITB Nigeria LTD and Hitech Construction Africa LTD Joint Venture

DOCUMENT REVISIONS

Rev	Date	Prepared by	Approved by	Remarks
00	12 Feb 2026	Tommaso Pagliani, ESG Officer	Carlos Silva, ESG Advisor	Draft Report

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1. PRESENTATION OF THE PROJECT AND THE PROPONENT

In the context of the Nigerian maritime industry, the Nigerian Port Authority (NPA) plays a pivotal role as a major organisation in a significant sector of the country's economy. In the evolving economic landscape, the Federal Government of Nigeria, through the NPA, is leveraging international financing from **UK Export Finance (UKEF)** and **Citibank** for the modernization of the Tin Can and Apapa Ports Project to enhance maritime safety, security, and compliance to globally acceptable standards. This initiative is crucial to enhancing the functionality and efficiency of the port complex, as the existing infrastructure is currently deteriorated and in need of rehabilitation.

Following approximately **47 years of operation**, the Quay Wall Structure has reached the expected material service life as it is showing signs of fatigue in the form of excessive settlement of slabs and deflection of the walls due to the corrosion of the steel sheet and erosion of the retained soil. These issues are currently impacting terminal operations across several berths, affecting port operability and limiting the capacity of cargo ships. Furthermore, the structural degradation poses safety risks for port activities, as the potential collapse of these elements could lead to accidents among terminal operator workers and damage to machinery.

The Tin Can and Apapa Ports Modernization Project aims to rehabilitate an existing quay wall length across **21 berths**, of approximately **4.56 kilometres** total, located between the southern shoreline of Tin Can Island and Apapa in Lagos, Nigeria. Construction will be executed by a **joint venture between Hitech Construction Africa Limited** and **ITB Nigeria Limited**. For the project activities, two temporary ancillary sites will be established: a General Support Yard of 4.8 hectares located northwest of Tin Can Island and a Laydown Area in Ibafon. These sites will support the project activities through the establishment of offices, a batching plant, storage, underground water supply, workshops and welfare facilities. At the quay wall, the construction site will cover an area of 40 to 50 metres wide from the shoreline, in sections of 200 metres across the quay wall footprint.

The Project footprint is located on land under the control of the NPA, which acts as landlord of the port estates and manages land used for port facilities (terminals, warehouses, administrative buildings and related infrastructure). The NPA’s mandate includes granting leases and concessions for port-related investment and development.

The construction period is expected to be **four (4) years**. The rehabilitated quay wall is designed for an approximate service life of **50 years**.

As part of the E&S Assessment process, the project developed an Environmental and Social Impact Assessment (ESIA) Study to comply with both international standards and Nigerian regulations. The project has been categorized as **Category B** as per the OECD project classification approach and **Category 2** as per the national EIA process in Nigeria.

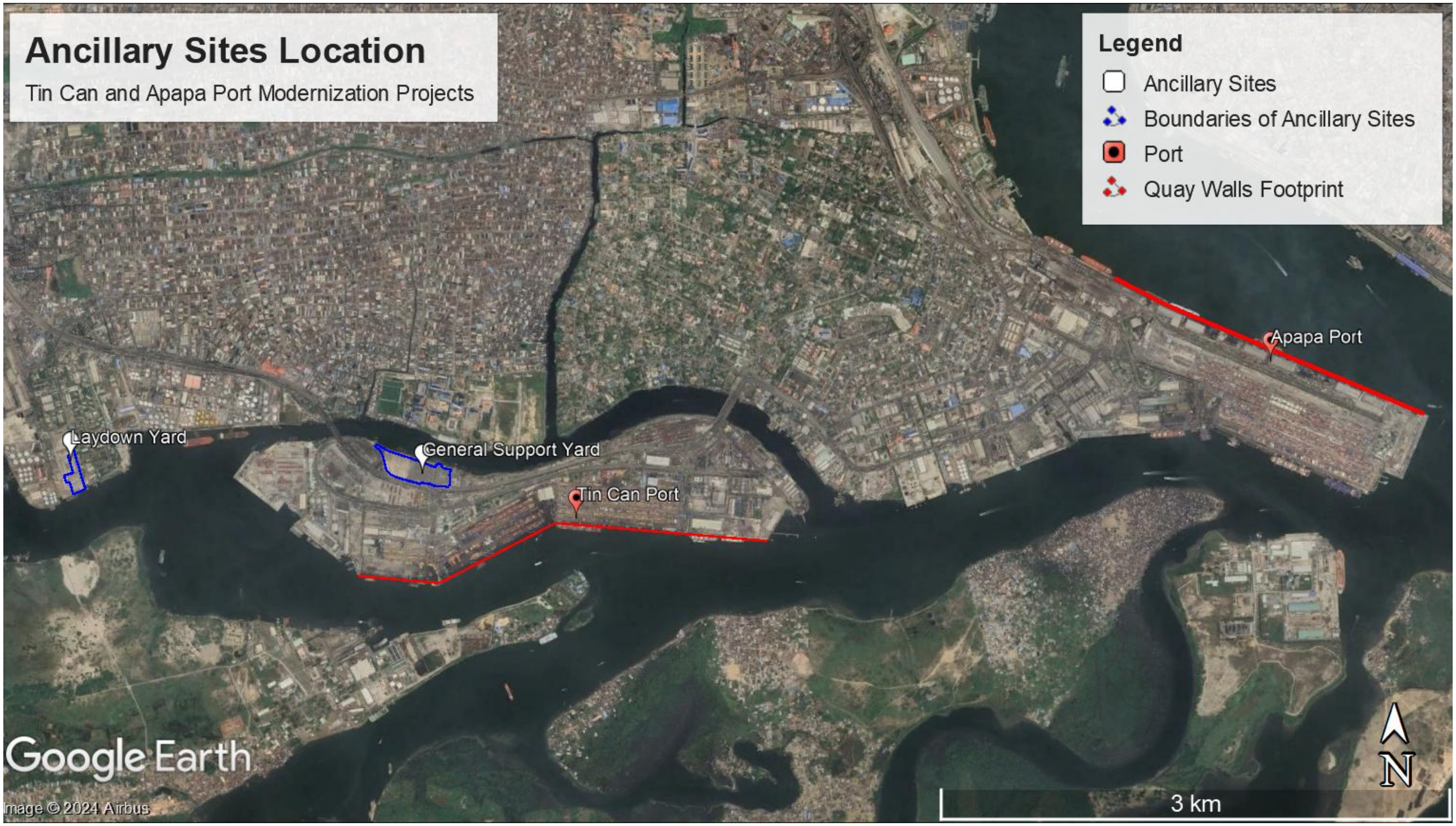


Figure 1: Location of Ports and Ancillary Sites

The following scheme presents the key elements of the ESIA process and NTS.

The ESIA Process
Natural Eco Capital and Multiple Development Services Ltd, the ESIA consultants, understands that NPA is planning to develop the rehabilitation of a quay wall at the Tin Can and Apapa Ports in Lagos. In this context, the project requires evaluating the environmental and social impacts and risk associated with the project and implementing mitigation measures to avoid adverse impacts. The project carried out the ESIA in line with the **IFC Performance Standards, Equator Principles IV, the World Bank EHS Guidelines** for development projects and the Nigerian regulations. The purpose of the ESIA report is to ensure that the environment is given full and proper consideration in the decision-making process with respect to potential activities having possible consequences on the environment. The ESIA report is used to predict, identify, assess, avoid, minimize, restore, and offset all the possible impacts of the planned activities and accidental events, so that environmental considerations can inform and be integrated into project decisions. It helps to select the best practicable location, layout, design, phasing, technologies, and products, to manage the impacts from activities and anticipate environmental restoration and offsetting, when needed. This ESIA report discusses the environmental, social and ecological baseline around the Tin Can and Apapa Ports Modernization Project and assesses the potential adverse and beneficial impacts that the project could have along with suitable mitigation measures to be part of an Environmental and Social Management Plan (ESMP) for the project. All this will be incorporated in an Environmental and Social Management System (ESMS) including all EHS policies, monitoring and implementation procedures, roles and responsibilities and reporting frequency as per the IFC PS1.

The Non-Technical Summary (NTS)
A non-technical summary (NTS) is a concise document that describes the ESIA process and its findings in a manner that is easily understood by the general public. The NTS supports public access to environmental and social information, consistent with applicable Nigerian ESIA disclosure requirements and international lender standards (including the IFC Performance Standards and the Equator Principles). It provides a description of the Project, a description of the measures envisaged to avoid, reduce and remedy significant adverse effects, the information required to identify and assess the main effects which the Project may have on the environment and an outline of the main Project alternatives.

According to available documentation, the Tin Can Island and Apapa Ports Modernisation Project ESIAs conclude that significant direct, indirect or cumulative adverse environmental and social impacts are unlikely, primarily because potential impacts are expected to be localised and temporary in nature and are to be managed through the mitigation and monitoring measures defined in the ESMP and the Contractor’s E&S management procedures. The ESIAs also include screening consistent with international lender requirements, including IFC Performance Standard 6, to identify any relevant

protected areas and critical habitat features within the Project's area of influence. Furthermore, given the Project's location within the existing Lagos port environment and its distance from international land and sea borders, no transboundary environmental impacts from the Project's planned activities are expected.

1.1 PROJECT ENTITIES

NIGERIAN PORTS AUTHORITY	**The Project Promoter, Buyer and Borrower:** The Nigerian Ports Authority (NPA) manages Nigeria's ports, including Lagos Port Complex and Tin Can Island Port, under the Federal Ministry of Transportation, overseeing modernization and development.
HITECH	**The Contractor:** The ITB - Hitech Joint Venture combines ITB Nigeria Limited's construction expertise, with Hitech Construction Africa Limited's regional prominence and commitment to international standards, supported by Dar Al-Handasah Consultants for technical design and engineering studies.
natural eco capital MDS	**The E&S Consultants:** A consortium, comprising Natural Eco Capital and Multiple Development Services Ltd, specialises in sustainability, environmental, and social safeguards for projects in Nigeria financed by institutions such as the World Bank, AfDB, and AFD, promoting sustainable practices and circular economy principles.
UK Export Finance citi	**The Lenders:** UK Export Finance (UKEF) is a UK government ministerial department and the UK's export credit agency (ECA). UKEF supports UK exports and international trade by providing guarantees and related support that enable exporters and project sponsors to access financing and manage payment risk. Citibank (Citi) acts as a Mandated Arranger for the Project.

2. PROJECT DESCRIPTION

2.1 Summary of Project Activities

The Tin Can and Apapa Ports Modernisation Project comprise the rehabilitation of existing quay walls at two sites in Lagos:

I. Apapa Port, where an existing quay wall length from berths 4 to 13 (approximately 2 km) will be rehabilitated,
II. Tin Can Island Port, where an existing quay wall length across 10 berths (approximately 2.5 km) will be rehabilitated, with both sites executed through a berth-by-berth system so that terminal operators can continue shipping activities in parallel with construction.

For each port, works are organised into construction sequences (each berth handed over before the next commences), with fenced construction corridors along the shoreline (Apapa typically 20–50 m wide depending on available space; Tin Can 40–50 m wide), implemented in sections across the quay wall footprint. The core rehabilitation sequence includes:

- Installation of guide walls.
- Installation of the new combi-wall system (steel casings and sheet piles).
- Demolition of the existing apron slab and removal of demolition materials.
- Backfilling and compaction to address soil loss/erosion behind the quay wall.
- Construction of rear barrettes and bored piles.
- Construction of the structural portal frame elements (including capping beams and transversal beams).
- Backfilling up to the required levels and casting of the new apron slab.
- Reinstatement/installation of berth facilities, followed by handover of each completed section.

Construction is supported by two temporary ancillary sites used for both port projects during construction phase: a General Support Yard (4.8 ha) in the northwest of Tin Can Island (with concrete batching plants, heavy equipment yard, repair workshops, project offices, storage areas and welfare facilities) and a Laydown Area in Ibafon (the "Savaric Yard", 1.5 ha) for laydown of combi-wall steel elements. The scope of work explicitly excludes capital or maintenance dredging; routine maintenance dredging of Lagos Harbour/access channels continues independently under NPA's statutory responsibilities via Lagos Channel Management Ltd and is not treated as an Associated Facility for the purposes of the ESIA.

2.2 Summary of Emissions, Discharges and Waste

Expected planned emissions, discharges and wastes from the Apapa and Tin Can Port Modernisation Projects during construction and operation include:

- Emissions to air from dust/particulates generated by material handling, batching plant operations, wind across unsurfaced materials and stockpiles, demolition and backfilling, and exhaust emissions from heavy machinery and diesel generators (including PM, NOx, SO_2 and CO);
- Noise emissions from construction activities (guide wall installation, demolition, backfilling, rear barrettes/bored piles, installation of berth facilities, generators, batching plants and vehicle traffic), including underwater noise during port activities and offshore transportation of materials;
- Wastewater from batching plant and yard facilities, managed via septic tanks during construction and collected for off-site disposal by a certified contractor;
- Solid wastes including general waste, medical waste, deteriorated PPE, oil filters and tyres, and construction wastes (excavation materials, demolition debris and packaging), managed under the Contractor's Waste Management Plan.

Workers are the primary receptors for air and noise impacts due to proximity to operating equipment, while residential receptors are located at greater distance and, given dispersion/attenuation and the industrial setting, impacts are expected to be negligible and within applicable limits. Air and noise will be managed through the relevant CESMP management plans (maintenance, low-sulphur diesel, minimising idling, PPE, scheduling and monitoring). Water quality risks relate mainly to runoff and potential spills during earthworks and shoreline works; land reclamation (approximately 2.8 m of fill within the existing quay alignment) is confined behind the new quay wall and does not expand the port footprint into the lagoon, and no dredging is included in scope. Solid and wastewater will be collected by Waste-Point Limited (LAWMA certified; ISWA membership number 15-1140) and disposed at the Olusoson Landfill in Lagos, with disposal arrangements subject to a Waste Audit scheduled for December and updates to the ESMP.

Following handover, routine port operations at Apapa Port and Tin Can Island Port will continue under the Nigerian Ports Authority and terminal operators. Expected planned operational emissions and wastes are typical of an active port environment and include: exhaust emissions from vessels at berth, cargo-handling equipment, trucks and standby generators (including NOx, SOx and PM, as well as greenhouse gases); fugitive dust where dry materials are handled and from yard movements; operational noise from cranes, yard equipment, generators and vehicle traffic; stormwater runoff from paved areas and maintenance yards that may contain oil/grease and suspended solids; sanitary wastewater from port facilities; and solid and hazardous wastes such as general waste, scrap materials, used oils and oily residues, contaminated rags/absorbents, batteries and spent filters. Ship-generated wastes and oily residues (where received) are managed through port waste reception and licensed disposal arrangements in line with applicable requirements.

Table 1: Estimated Waste Generation from construction activities

Waste Description	Waste Quantity
Steel Pile Excavation Spoils – 41.4 m^3 per 795 Piles with 15% Increase	37,850 m^3
Concrete Pile Excavation Spoils – 27.1 m^3 per 1220 Piles with 15% Increase	38,021 m^3
Rear Barrette Excavation Spoils – 83.1 m^3 per 610 Piles with 15% Increase	58,294 m^3
Concrete Demolition of Existing Quay Slab	23,040 m^3
Steel Reinforcement Waste – Estimated at 3% of Total Weight Required	1,403 Tons
Concrete Waste – Estimated at 1% of Total Volume Required	2,420 m^3
Workforce Sewage Generation	160 m^3 per Day
Workforce Waste Generation – Average 0.15 kg/person per day	217 kg per Day
PPE Kits – Helmets, Safety Boots, Reflective Vests, Goggles, etc.	2 Kits per Year per Person

2.3 Project Alternatives

2.3.1 No-Go Alternative (Zero Alternative)

The "No-Go" (or "zero alternative") describes the likely situation if the Apapa and Tin Can quay wall rehabilitation is not undertaken. Without rehabilitation, the continued deterioration of the existing quay wall structures would persist, with progressive settlement of the apron slabs and further outward deflection of the quay walls driven by corrosion of the steel elements and erosion of retained soils. This would increase safety risks for port workers and equipment and could lead to partial or full failure of quay wall sections. In practical terms, this would further constrain terminal operations, reduce the ability of berths to safely accommodate cargo ships, and undermine the operational efficiency and competitiveness of both port complexes, with knock-on implications for vessel traffic, cargo throughput, and future investment and development. On this basis, the No-Go alternative is not considered feasible from an operational and safety perspective.

2.3.2 Alternatives Considered and Selected Approach

The ESIA includes a qualitative review of alternatives. At an early stage, the Contractor reviewed two quay wall design options prepared by Dar. These options differed mainly in the structural configuration of quay wall elements and did not introduce material differences in environmental or social risk. For this reason, they were not assessed further in the ESIA, and the alternatives assessment focused instead on options for the pre-construction and use of ancillary sites. Two options were considered:

I. establishing a single ancillary site at the General Support Yard on Tin Can Island, which offers proximity and transport efficiency but presents constraints including limited storage area, higher rental costs, the need for substantial pre-construction works (utilities, drainage, facilities and septic tanks), and sensitivity considerations due to nearby receptors and ecosystem services; and

II. relying solely on the Savaric Yard, which is already equipped with drainage, septic tanks, generators, boreholes and water treatment and includes a jetty suitable for offshore transport of large structures, but would increase road transport distance and traffic-related constraints, potentially affecting construction efficiency and concrete quality during long transport in congested conditions, with additional sensitivity due to adjacent residential areas and potential aquatic impacts associated with offshore transport.

Based on this analysis, the Project selected a combined approach using both locations: Tin Can Island as the General Support Yard to maximise efficiency for day-to-day support functions due to proximity to the construction areas, and the Savaric Yard to store and enable offshore transport of large structures via its jetty, thereby balancing efficiency, cost considerations, and the minimisation of environmental and social impacts.

3. BASELINE CHARACTERISTICS: DESCRIPTION OF THE EXISTING ENVIRONMENT

The objective of this chapter is to provide an in-depth understanding of the physical, biological, socio-economic, and cultural heritage aspects of the environment for the Apapa Port and Tin Can Island Port modernisation works. This information helped to create a comprehensive profile of the project areas and provided relevant data on physical, biological, archaeological, and socio-economic conditions through literature and field investigation. As a result, the environmental and social baseline conditions have been described with a level of detail that allows for a precise assessment of the various environmental and social impacts before the implementation of the project interventions. The Baseline Study provides the environmental and social status of the port locations, ancillary sites, and surrounding areas, supporting a proactive approach to protect the environment and local communities' well-being while enabling a sustainable port that benefits both the economy and the environment. The main study area concerns the strict perimeter of the Project, including the construction areas at the quay walls, the ancillary sites, and the route to be used during offshore transportation of structures. For the ESIA themes addressed in this study, an Area of Influence (AOI) has been defined beyond the Project footprint, based on expert judgement and experience and in line with a component-based approach. The AOI comprises: (i) the quay wall work areas within the Apapa Port and Environs Area and the Tin Can Island Port and Environs Area; (ii) the General Support Yard and Laydown Yard; and (iii) the principal access roads and marine routes used for movement of materials, equipment and workforce. For each theme (e.g., air quality, noise, traffic, community health and safety), the ESIA applies defined distances/extent from these components to capture the relevant receptors and pathways.

3.1 Physical Environment

3.1.1 Climate

Nigeria spans three broad climatic zones: humid tropical conditions in the south, tropical savannah across much of the central region, and hot semi-arid Sahelian conditions in the far north, with rainfall generally decreasing from south to north. The Tin Can and Apapa Ports Modernization Project is located in Lagos State on the Atlantic coast, within a humid tropical coastal climate influenced by proximity to the ocean and seasonal winds. The climate is typically described by a wet season dominated by moist south-westerly air from the Atlantic and a drier season influenced by north-easterly Harmattan conditions. Temperatures remain consistently warm (generally above 20°C) and relative humidity is high; according to available documentation, the annual mean relative humidity around Lagos is approximately 88%.

The main weather-related hazards relevant to the Lagos coastal environment are heavy rainfall events, squalls, and associated localised flooding and operational disruption. Tropical cyclones are not considered a typical hazard along the Nigerian coast, with historically very low cyclone activity and no recorded direct hurricane/typhoon landfalls; on this basis, cyclone risk for the Project area is considered low. No site-specific, threshold-based sampling or compliance testing is applicable for this climate baseline description.

3.1.2 Rainfall

Rainfall occurs throughout the year in Lagos, with a clear wet–dry seasonal pattern driven by the seasonal movement of the Inter-Tropical Discontinuity (ITD) and associated monsoon winds. For the Tin Can and Apapa Ports Modernization Project area, rains typically begin in late April and continue until around October, with August often showing a temporary reduction in rainfall (“August break”).

According to available documentation, NiMET records for 1993–2023 indicate average monthly rainfall is lowest in January and December (around 14 mm) and peaks in June (around 287 mm). The ESIA also references 30 years of hourly weather model simulations and observed historical climate data that corroborate this wet and dry season pattern. No threshold-based compliance criteria apply to rainfall baseline reporting in the NTS.

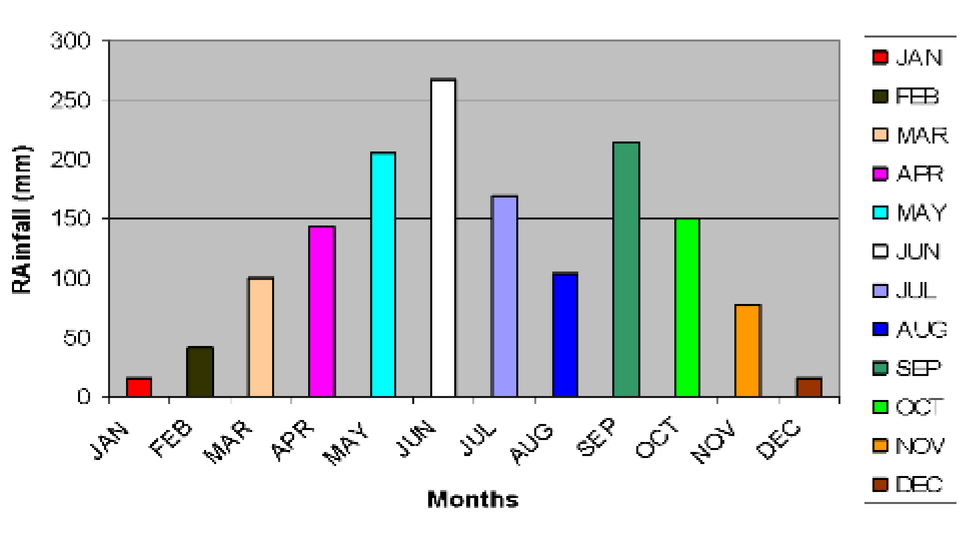


Figure 2: Average Rainfall in Lagos for the period 1993-2023. Source: NIMET, 2023

3.1.3 Ambient Air, Noise and Vibrations

Baseline ambient air quality for the Tin Can and Apapa Ports Modernization Project was characterised using a combination of desktop information and field measurements covering common "classical" pollutants and particulate matter (CO, CO_2, NO/NO_2, SO_2, VOCs, $PM_{2.5}$ and PM_{10}). Desktop-based data indicate generally fair to excellent ambient air quality levels for key pollutants in the wider Lagos port area (e.g., $PM_{2.5}$ 12 µg/m³; PM_{10} 31 µg/m³; O_3 52 µg/m³; CO 181 µg/m³; NO_2 1 µg/m³; SO_2 0 µg/m³). According to available documentation, field monitoring was undertaken during a transitional season at representative port locations and nearby sensitive receptors, with measurements taken across four daily periods and averaged to a 24-hour value. The ESIAs report that gaseous pollutants were within applicable limits and that $PM_{2.5}$ and PM_{10} values at monitored points were within the referenced IFC/WHO guideline values of 25 µg/m³ ($PM_{2.5}$) and 50 µg/m³ (PM_{10}).

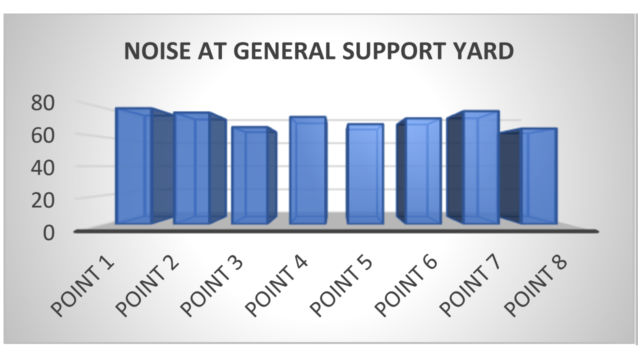
Pollutant	Description	Concentration
$PM_{2.5}$ Fair	Fine Particulate Matter are inhalable pollutant particles with a diameter less than 2.5 micrometres that can enter the lungs and bloodstream, resulting in serious health issues. The most severe impacts are on the lungs and heart. Exposure can result in coughing or difficulty breathing, aggravated asthma, and the development of chronic respiratory disease.	12 µg/m³ **AQI**[1]: 42
PM_{10} Fair	Particulate Matter are inhalable pollutant particles with a diameter less than 10 micrometres. Particles that are larger than 2.5 micrometres can be deposited in airways, resulting in health issues. Exposure can result in eye and throat irritation, coughing or difficulty breathing, and aggravated	31 µg/m³ **AQI:** 36

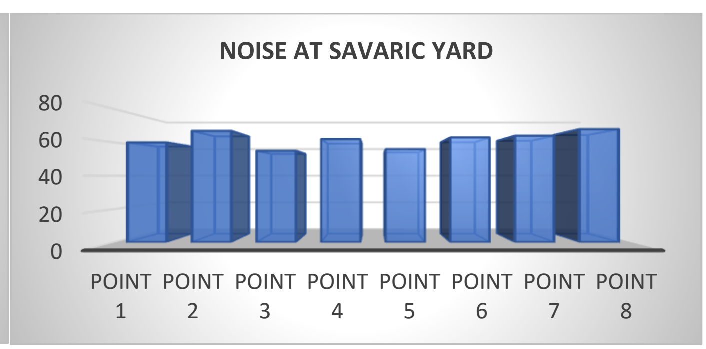
[1] The Air Quality Index (AQI) is internationally recognized and used by organizations such as the World Health Organization (WHO) and environmental agencies worldwide to assess and communicate the health risks associated with varying levels of air pollution, aiding in informed decision-making and policy development.

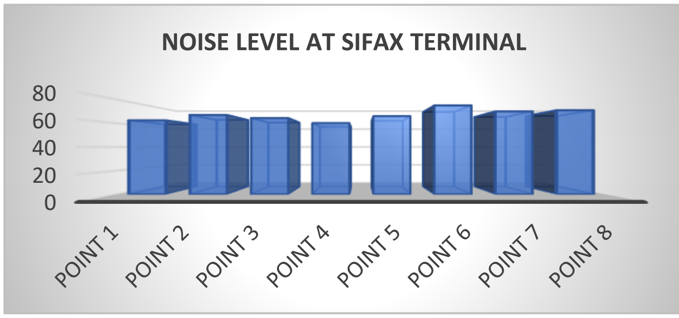
Pollutant	Description	Concentration
	asthma. More frequent and excessive exposure can result in more serious health effects.	
O_3 Fair	Ground-level Ozone can aggravate existing respiratory diseases and also lead to throat irritation, headaches, and chest pain.	52 µg/m³ **AQI:** 17
CO Excellent	Carbon Monoxide is a colourless and odourless gas and when inhaled at high levels can cause headache, nausea, dizziness, and vomiting. Repeated long-term exposure can lead to heart disease	181 µg/m³ **AQI:** 2
NO_2 Excellent	Breathing in high levels of Nitrogen Dioxide increases the risk of respiratory problems. Coughing and difficulty breathing are common and more serious health issues such as respiratory infections can occur with longer exposure.	1 µg/m³ **AQI:** 1
SO_2 Excellent	Exposure to Sulphur Dioxide can lead to throat and eye irritation and aggravate asthma as well as chronic bronchitis.	0 µg/m³ **AQI:** 0

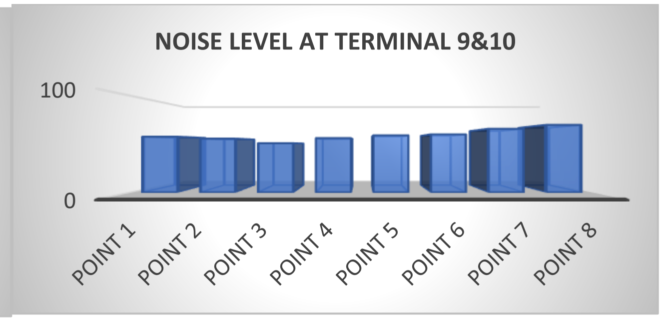
Air pollution is a critical issue in Nigeria, especially in industrialized zones. Despite these generally favourable levels, it's imperative to continue caution and vigilance during construction activities to ensure health and safety risks of workers are minimized, as even minimal exposure to PM2.5 can pose serious health risks.

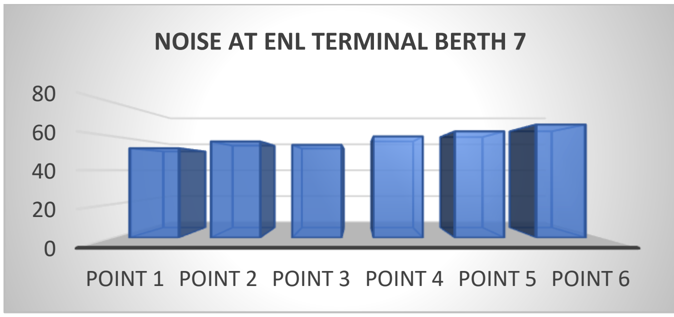
Baseline noise levels were assessed in an operational port environment (i.e., existing activities such as loading/unloading, traffic and equipment operation). Measurements were taken at representative locations across the Project's key work areas and nearby receptors using a calibrated sound level meter (94 dB calibration check), during four one-hour periods (morning, midday, evening and midnight), and averaged to a 24-hour value. Monitoring locations included, at Tin Can Island Port, the General Support Yard, Savaric Yard, Sifax Terminal and Terminal Berth 9, and at Apapa Port, the ABTL Terminal (Berth 2). According to available documentation, the ESIA reports that baseline noise levels at monitored locations were within the applicable Federal Ministry of Environment limits for port/industrial environments, including relevant 85–90 dB(A) occupational benchmarks over an 8-hour working period where applicable.

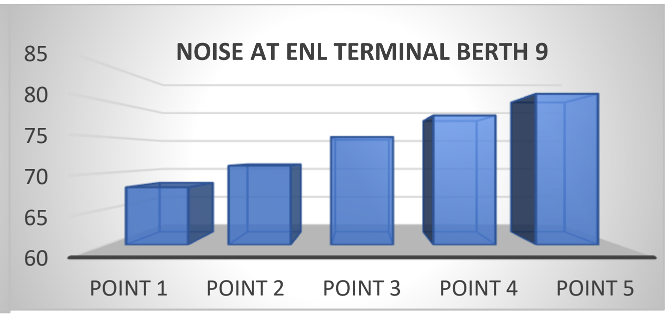


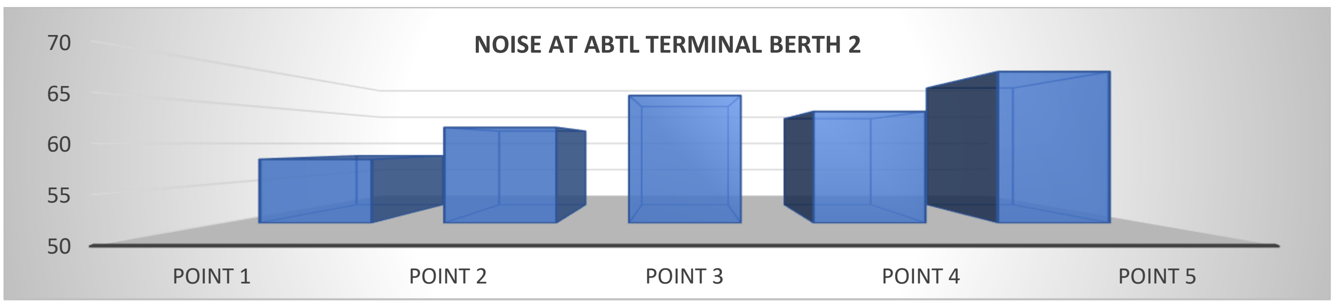












Underwater vibration was measured at selected water locations around the port and at control points (including Marina locations) using a vibration meter, with surface water quality also measured at the same locations to provide supporting context. According to available documentation, this approach is aligned with the IFC/World Bank EHS Guidelines for Ports, Harbors, and Terminals, recognising that elevated underwater vibration can affect aquatic habitats and species. Baseline results indicate low vibration values at control points and measurable but variable vibration levels at port-proximate locations, providing a reference condition against which any changes associated with construction activities and vessel movements can be assessed.

LOCATION	GPS	TEMPERATURE (°C)	VIBRATION (m/s^2)
Ito-agun	Lat 6.431315° Long 3.351945°	41°C	$0.33 m/s^2$
Sabokoju	Lat 6.431299° Long 3.354528°	40°C	$0.39 m/s^2$
Eco Support	Lat 6.434602° Long 3.389848°	36°C	$0.28 m/s^2$
Apapa Port (1)	Lat 6.433038° Long 3.376358°	37°C	$0.35 m/s^2$

Apapa Port (2)	Lat 6.436334° Long 3.384151°	34°C	0.27m/s^2
Apapa Port (3)	Lat 6.434916° Long 3.389608°	35°C	0.32m/s^2
Marina (1)	Lat 6.439632° Long 3.394722°	34°C	0.01m/s^2
Marina (2)	Lat 6.445217° Long 3.389792°	35°C	0.02m/s^2
Marina (3)	Lat 6.445474° Long 3.389865°	36°C	0.04m/s^2

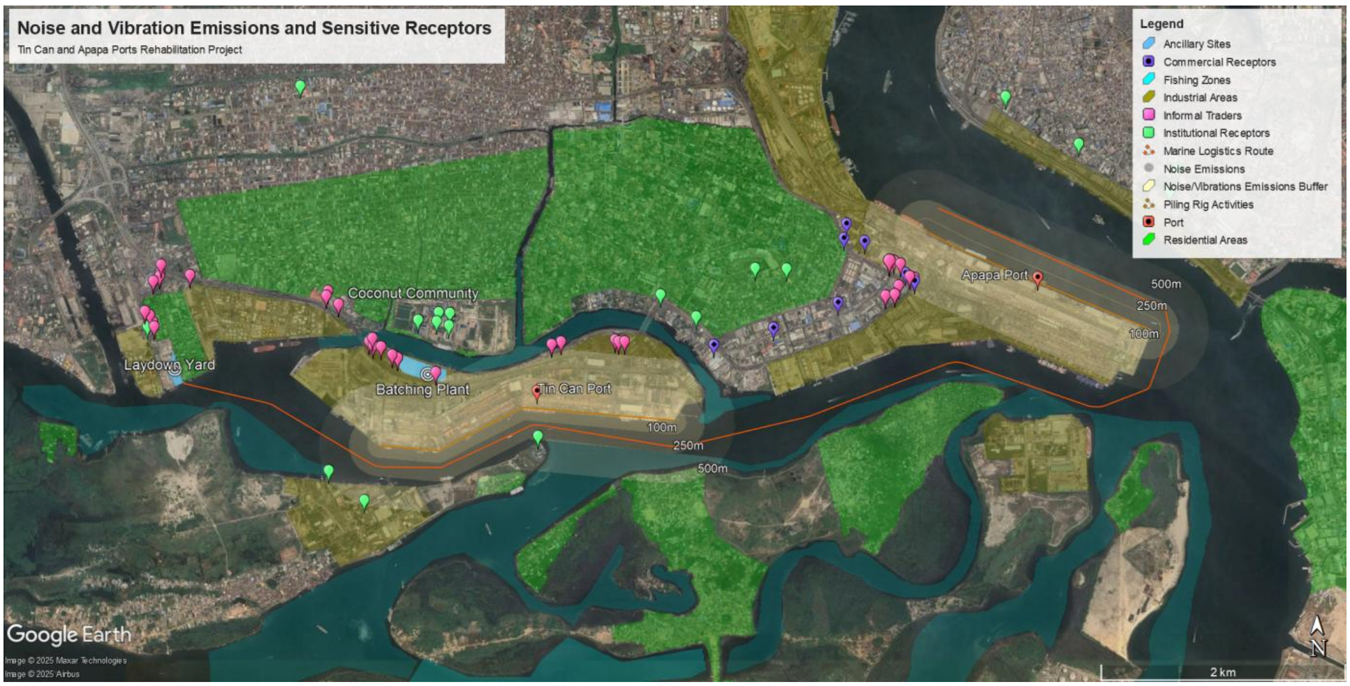


Figure 3: Noise and Vibration Emissions and Sensitive Receptors

Impact Source	Risks	Primary Receptors	Typical Distance to Receptors	Risk Level	Notes / Rationale (Concise)
Piling, sheet-pile driving, concrete breaking (quay wall).	Airborne noise; low-frequency vibration traveling through water and quay structures	• Terminal workers • Boat operators • Fishermen • Informal traders near port gates • Residential pockets (Coconut, Ibafon)	• Adjacent terminal areas: 10–50 m • Boat users: 15–100 m from piling fronts • Residential pockets: 250–700 m	**Moderate**	Noise propagates efficiently across open waterfront; vibration detectable in adjacent terminals, with residential receptors located beyond the near-field vibration zone
Heavy equipment, cranes, excavators, generators	Localized airborne noise; vibration through slab	• Terminal workers • Truck drivers • Port gate users	• Within quay: 10–30 m • Gate areas: 200–400 m	**Moderate**	Noise dissipates inland; highest risk to workers and people near quay edges.
Batching plant, material handling, equipment yard (Savaric + General Support Yard)	Dust (PM10/PM2.5) via wind; localized noise	• Workers • Adjacent residential cluster near Ibafon • Vendors at yard entrances	• Workers: 0–20 m • Ibafon residential: 80–150 m	**Moderate**	Dust depends on wind direction; manageable with watering and enclosure.
Truck movements on *internal port access roads and gate approaches* (both ports)	Traffic congestion; risk of collision with community; noise	• Informal traders • Pedestrians • Transport workers • Motorbike taxis	Pedestrian/trader zones often within **0–5 m** of traffic corridors	**Moderate**	High mixing of pedestrians, traders, trucks; predictable and manageable with controls.
Barge/tugboat traffic along Badagry Creek	Collision risk; wake turbulence; restricted visibility	• Artisanal fishermen • Informal passenger boats • Small-scale transport vessels	Typical overlap zones **20–80 m** from barge routes	**Moderate–High**	Most sensitive marine interface; requires strict coordination and exclusion zones.
Marine exclusion zones and temporary access restrictions	Community entering exclusion areas (unaware or intentionally)	• Fishermen • Informal boat operators	Exclusion zones radius **50–150 m**	**Moderate**	Risk depends on clarity of communication and marking of zones.
Hazardous materials (fuel, lubricants, concrete washout)	Spills reaching public areas or waterways	• Fishermen • Waterfront users • Informal traders • Terminal staff	Marine refueling area to fishing paths: **50–200 m**	**Moderate–High**	Marine spills pose direct community livelihood impact; low-probability land spills manageable with containment
Construction waste, debris, scrap metal	Public exposure; cuts; blocked pathways	• Workers • Traders • Pedestrians near site boundaries	Boundary proximity **0–20 m**	**Low–Moderate**	Easily controlled with fencing; risk increases when fencing is breached.
Security interface (access control, crowding)	Miscommunication; aggressive enforcement; crowding at gates	• Truck drivers • Informal traders • Community members	Gate crowding zones **0–5 m**	**Moderate**	Existing VPSHR reduces severity; risk linked to congestion periods.
Labor influx and health exposure	Respiratory diseases, interactions at markets/terminals	• Local communities • Vendors • Transport workers	Worker-community interface areas **0–10 m**	**Low–Moderate**	Manageable with awareness, hygiene, and health coordination.

3.1.4 Geology

The Tin Can and Apapa Ports Modernization Project is located within Lagos' low-lying coastal and lagoonal environment, characterised by creeks, lagoons and intertidal conditions. The underlying geology comprises recent alluvial deposits and coastal plain sands, within a broader regional stratigraphy that includes the Coastal Plain Sands (Benin Formation) and deeper sedimentary units of the Abeokuta Group. Topography in the wider Lagos area is generally flat and close to sea level (average elevation reported at ~5 m). The ESIAs describe the setting as geologically stable, with no tsunami or earthquake hazards identified for the project area, while noting that coastal dynamics and erosion processes influence the wider barrier–lagoon system.

Subsurface conditions in the port environment are described as layered sequences of clays, silts and sands, consistent with a coastal plain and lagoonal depositional setting. This provides the geological context for ground conditions relevant to quay infrastructure and works undertaken within existing port footprints.

Soils in the project area are described as water-saturated or frequently inundated sandy soils typical of coastal/lagoon margins, with wider Lagos coastal soils commonly classified as Arenosols. A baseline soil survey was reportedly undertaken within the area of influence, with laboratory analysis by LASEPA. Results indicate sand-dominant soils, low organic matter and near-neutral pH. The ESIA also reports that measured indicators for metals and hydrocarbons were generally low, with oil/grease detected at minimal levels and remaining within the referenced thresholds for the assessment.

3.1.5 Hydrology and water resources

The Project is located within a connected creek–lagoon–ocean system that supports navigation and receives runoff from densely industrial and residential areas. This hydrological connectivity means that local changes in water quality can propagate beyond the immediate works footprint, particularly through tidal exchange and stormwater pathways.

According to available documentation and baseline surveys referenced in the ESIA, surface water quality in the wider area of influence is already degraded, with elevated turbidity and indicators consistent with organic pollution, and hydrocarbon signatures reported in several sampling locations. Groundwater conditions are also described as sensitive in coastal Lagos, where shallow aquifers can be vulnerable to contamination pathways and saline intrusion. Where boreholes are used for supply, the ESIA indicates that suitability should be confirmed through testing, including comparison against applicable water quality standards; parameter-by-parameter compliance status is presented in the ESIA datasets rather than in the NTS.

Overall, the baseline establishes that water resources are already under pressure and provides the reference point against which any Project-related changes in turbidity, hydrocarbons and related parameters are assessed. For detailed sampling locations, laboratory results, and explicit comparison to regulatory and international thresholds, the ESIA should be consulted.

3.1.6 Hydrography

According to available documentation referenced in the ESIA, tidal ranges along the Lagos coastline are modest and reduce inland through the lagoon and creek system. Spring tides are reported at approximately 1.0 m at Lagos Bar, decreasing to around 0.8 m in Badagry Creek, with typical tidal levels referenced to Chart Datum (0.0 m) including MHWS of about +1.0 m, MHWN +0.8 m, MSL +0.6 m, MLWN +0.4 m and MLWS +0.2 m. The ESIA notes that tide levels can be influenced by wet-season river discharge.

Current measurements and hydrodynamic modelling (including ADCP-based data) indicate generally weak flows under neap conditions and higher velocities during spring tides. Reported surface currents are commonly around 0.6 m/s, with modelled peak surface flows reaching approximately 0.9 m/s near the eastern end of Tin Can Island; near-bed currents are lower, typically up to around 0.5–0.6 m/s. The ESIA describes flow patterns as complex due to lagoon–creek interactions and potential lag between water level changes and peak current timing.

Wave conditions are described as dominated by south to south-westerly swell and locally generated wind waves, with nearshore refraction influencing circulation and sediment mobility. This hydrographic context informs how suspended sediments, potential contaminants, and floating debris can be transported within and beyond the immediate port areas, with further detail provided in the ESIA modelling outputs.

3.2 Biological Environment

This chapter establishes the biological baseline for the Project Area of Influence (AOI) to support lender-aligned biodiversity risk screening and impact management consistent with IFC Performance Standard 6. According to available documentation, the baseline combines desk-based review of open-source biodiversity datasets (e.g., IBAT, IUCN Red List, BirdLife International, Global Forest Watch), targeted stakeholder consultations, and wet- and dry-season field surveys across key taxonomic groups, with laboratory verification where required. Nigeria has a total of 1,001 protected areas, largely Forest Reserves; however, desktop-based screening did not identify any protected areas within a 50 km radius of the Project area. The outputs are used to characterise habitat condition (modified/natural) and to inform the critical habitat screening and the application of the mitigation hierarchy within the ESIA.

3.2.1 Fauna and Flora

Nigeria has high biodiversity at a national scale; however, the Tin Can and Apapa Ports Modernization Project is situated in a long-established urban–industrial setting where habitats are highly transformed. According to available documentation, the Area of Influence comprises remnant and generally degraded mangrove and wetland patches, secondary vegetation/regrowth, and disturbed wastelands, reflecting historical land take and fragmentation.

The flora survey recorded 35 herbaceous species across 15 families, with Poaceae and Cyperaceae most represented. Species composition is dominated by disturbance-tolerant grasses and sedges (notably Cynodon dactylon and Rhynchospora corymbosa). The ESIA reports that recorded plant species are largely classified as Least Concern or Not Evaluated, with no flora species identified as being of national conservation importance within the surveyed area.

Faunal observations and records are similarly typical of modified coastal/urban mosaics, with baseline listings dominated by Least Concern species across birds, reptiles, amphibians and small mammals. The ESIA notes limited indications of higher-sensitivity taxa, including an isolated Vulnerable record for Tawny Eagle (Aquila rapax) and Data Deficient/Not Evaluated statuses for certain invertebrates and fish.

Aquatic ecology in the Area of Influence is linked to the brackish Lagos Lagoon system and supports fishing-based livelihoods, with seasonal shifts in fish assemblages between wet and dry seasons. According to available documentation, baseline conditions are already influenced by wider lagoon stressors—urban runoff, wastewater inputs, and ongoing dredging/sand mining—which are not created by the Project but form the background condition in which port activities take place; increased activity could exacerbate these pressures if spills, runoff, or waste controls are ineffective.

The ESIA also records **low-density water hyacinth (Eichhornia crassipes)** in the wider water bodies. Water hyacinth is an invasive aquatic plant that spreads rapidly by producing “daughter” plants, forming dense mats that can obstruct navigation, reduce light penetration, crowd out native species, and lower dissolved oxygen. In Nigeria and neighbouring countries it is commonly present and can expand quickly, with mats reported to double in size within approximately **6–18 days** under favourable conditions. While complete eradication is generally not feasible once established, baseline identification at low density is relevant because it provides an early reference point for monitoring and management. The ESIA indicates that management measures will be developed with guidance from the **Nigerian Ports Authority (NPA)**, recognising the need to prevent escalation that could affect navigation and aquatic habitat quality.

Figure 4: Common Water Hyacinth - Eichhornia crassipes

The fish diversity of Lagos Lagoon demonstrates clear seasonal patterns. During the wet season, the influx of freshwater supports a richer diversity of freshwater species, as noted in the increased presence of cichlids, catfish, and other species adapted to brackish and freshwater environments. Conversely, the dry season supports a higher concentration of marine species, such as Lutjanidae and Carangidae, as the salinity of the water increases with reduced freshwater inflows.

The Lagos Lagoon system exhibits significant spatial and temporal dynamics, attracting both freshwater and marine fish species, which periodically inhabit the lagoon in search of food and nursery grounds. The results of the survey highlight this seasonal variation, where fish species from both origins demonstrate differing population patterns across wet and dry seasons. The presence of key families such as Mugilidae, Cichlidae, Clupeidae, Bagridae, Sphyraenidae, Pomadasydae, and Lutjanidae across both seasons, as recorded in this study, confirms their classification as permanent inhabitants of the Lagos Lagoon ecosystem.

The findings also revealed that a higher diversity of marine-origin fish species was recorded during the dry season. This seasonal shift can be attributed to the reduced salinity of the lagoon during the dry season, which creates favorable conditions for juvenile marine species to migrate and inhabit the lagoon. These results align with previous studies (Fagade and Olaniyan, 1974; Amadi, 1990; Ayoola and Kuton, 2017), which indicated that the juvenile stages of many marine species thrive in lower salinity waters such as those found in the Lagos Lagoon during the dry season.

However, the lower richness of fish species observed in this study compared to previous research may indicate an ongoing decline in biodiversity, likely influenced by human activities such as dredging and sand mining. These activities disrupt natural breeding and spawning grounds for several fish species, hindering their movement and migration patterns between marine, freshwater, and brackish environments. Additionally, the practice of sand filling further exacerbates these challenges, inhibiting the movement of fish species and disrupting ecological balance.

The continued discharge of untreated wastewater and other pollutants into the lagoon is also a pressing concern. The pollution threatens the delicate ecological equilibrium, further reducing fish population dynamics and overall biodiversity. The degradation of the lagoon due to these harmful practices may explain the observed decline in annual fish production, a concern for both local livelihoods and broader food security.

The Lagos State government's proposal to introduce cage culture in the lagoon as a means to enhance food security is timely and could provide a sustainable solution to supplement fish stocks. Given the importance of fish and fisheries products to the local diet—offering a source of high-quality protein, essential nutrients, and Omega 3 fatty acids—efforts to preserve the lagoon's fish population should be a priority.


Figure 5: Sarotherodon melanotheron


Figure 6: Chrysichthys nigrodigitatu


Figure 7: Chrysichthys nigrodigitatus


Figure 8: Pomadasys jubelini


Figure 9: Oreochromis niloticus


Figure 10: Callinectes amnicola


Figure 11: Heterotis niloticus


Figure 12: Parapenaeopsis atlantica

3.2.2 Critical Habitat Assessment

A Critical Habitat Assessment is a targeted screening undertaken with reference to IFC Performance Standard 6 to determine whether the Tin Can and Apapa Ports Modernization Project is located in, or could affect, areas of exceptional biodiversity value (for example habitats supporting threatened species, key migratory concentrations, or unique ecosystems). The assessment informs whether the IFC PS6 “Critical Habitat” definition is triggered for the Project.

For the Project, the assessment was based on a desktop review of available biodiversity and conservation information and on the hydrobiological and wildlife studies completed as part of the ESIA. According to available documentation, Nigeria has 1,001 protected areas (predominantly Forest Reserves); however, desktop screening did not identify protected areas within a 50 km radius of the Project areas.

Overall, the ESIA concludes that the Project areas are highly modified urban/port environments and do not meet the IFC PS6 definition of Critical Habitat. On this basis, the ESIA indicates that the Project is not expected to affect internationally important biodiversity features, and biodiversity risk management is addressed through the Project’s standard ESMP framework (including pollution prevention, waste management, and monitoring) as presented in the ESIA.

3.3 Social Environment

3.3.1 Administrative Overview

The Tin Can and Apapa Ports Modernization Project is located in Lagos State within the Apapa port and industrial district of metropolitan Lagos. The Project covers two existing port estates, Apapa Port and the Tin Can Island Port Complex (TCIPC), within a coastal, highly urbanised setting characterised by intensive logistics and industrial land use. According to available documentation, the development of this district has been strongly driven by port activity and trade-related industrialisation, alongside persistent constraints including chronic traffic congestion and associated air/noise emissions and pressure on local infrastructure around the port estate.

Lagos State operates a multi-tier governance structure comprising State-level administration, 20 Local Government Areas (LGAs) and Local Council Development Areas (LCDAs) that support local service delivery (including local roads, sanitation, waste management and primary healthcare). The port zone interfaces with densely populated and commercially active neighbourhoods and transport nodes, including areas within Apapa, and adjacent LGAs such as Ajeromi-Ifelodun and Amuwo-Odofin, as well as activity centres linked to Lagos Island, all of which influence local mobility patterns and community–port interactions.

While the ports are physically located within Lagos State, **Nigeria’s ports are federally owned and regulated, with the Nigerian Ports Authority (NPA) acting as the principal port manager under a landlord model with terminal operators**. Other public agencies typically present in port areas include customs, maritime administration and inland waterways authorities. This institutional setting defines the core administrative context relevant to permitting, oversight and stakeholder coordination for the Project.

For the socio-economic baseline, the ESIA defined an Area of Influence (AOI) extending to an 800 m buffer around the Project area to capture likely direct and indirect interactions associated with port modernisation, including traffic, noise, air emissions and community interface considerations.

3.3.2 Demography

The Tin Can and Apapa Ports Modernization Project is located in metropolitan Lagos, Nigeria's largest urban and economic centre, with a highly diverse population and sustained in-migration linked to port activity, trade and industrial employment. According to available documentation, Lagos State population growth is projected at approximately 3.5% per annum, implying continued pressure on housing, transport networks and public services in port-adjacent areas.

Socio-economic conditions vary materially across Lagos. The available documentation characterises poverty using multidimensional indicators and reports that less than 30% of the Lagos population is multidimensionally poor. The same documentation notes that youth and women may remain disproportionately exposed to vulnerability drivers, including education attainment, skills development and access to stable employment.

For the areas most directly connected to the ports, the ESIA references Lagos State Bureau of Statistics (LBS) estimates for key LGAs that host or border the Project area, including Apapa, Ajeromi-Ifelodun, Amuwo-Odofin and Lagos Island. Within the wider Area of Influence, communities range from established residential and commercial zones to dense informal settlements and waterside communities, with differences in basic infrastructure and service access. This variability is relevant for understanding the distribution of sensitive receptors and the baseline exposure of communities to existing pressures around the port and transport corridors.

3.3.3 Land Ownership and Planning

Land tenure in Nigeria is governed primarily by the Land Use Act (1978), under which all land within each State is vested in the State Governor, held in trust for the common benefit of Nigerians. In practice, individuals, communities and entities typically hold rights of occupancy (statutory or customary) rather than absolute ownership, with formal tenure commonly evidenced through instruments such as a Right of Occupancy or Certificate of Occupancy (C of O) issued by the competent State authority.

Within this national framework, tenure arrangements may include state-administered statutory tenure, individual or corporate rights of occupancy, customary tenure (particularly relevant to community lands), and leasehold arrangements. While customary rights may be recognised in Lagos State, legal security is generally strengthened where landholding is formalised through a C of O. According to available documentation, the Land Use Act is frequently cited as creating administrative complexity and longer timelines for formal land processes.

For the Tin Can and Apapa Ports Modernization Project, the Project footprint is understood to be located on land under the control of the Nigerian Ports Authority (NPA), which acts as landlord of the port estates and manages land used for port facilities (terminals, warehouses, administrative buildings and related infrastructure). The NPA's mandate includes granting leases and concessions for port-related investment and development. According to available documentation, no land ownership disputes have been identified within the Project areas.

From the ESIA baseline perspective, the principal land and planning feature is therefore the NPA-controlled port estate, alongside adjacent communities where customary landholding may exist but remains subject to Lagos State administration under the Land Use Act, with formalisation typically evidenced through a C of O.

3.3.4 Housing

The Tin Can and Apapa Ports Modernization Project is situated in a highly urbanised coastal context where housing conditions vary materially across the Project's Area of Influence. In port-adjacent mainland areas and along key access corridors, housing is predominantly formal and dense, reflecting long-established port, logistics and commercial land use. In parallel, parts of the wider lagoon environment include semi-urban and village settings where informal waterfront settlements are present.

In and around Lagos Island, the built environment includes a wide range of typologies, including older building stock and mixed residential–commercial uses typical of a major port-city economy. According to available documentation, housing availability and affordability pressures in central Lagos have contributed to the conversion of some residential buildings to commercial use, with knock-on effects on commuting patterns and demand in peri-urban residential areas.

Within the wider Project area, particularly in lagoon-side communities such as Ilashe, Snake Island and Ogogoro, housing is often characterised by informal waterfront construction. According to available documentation and field observations, dwellings may be built over water or on stilts, using locally available materials (e.g., wood, bamboo and corrugated metal sheets, with some use of concrete blocks). Elevated foundations and raised platforms are commonly used to manage waterlogged ground conditions and flood/tidal exposure, and these settlements are typically more constrained in access to formal infrastructure and public services.

Overall, baseline housing conditions indicate that the Project's social receptors include both high-density formal neighbourhoods and more vulnerable informal waterfront communities, with differing sensitivities to nuisance effects and environmental stressors in a heavily industrial and transport-constrained port setting.

3.3.5 Services

The Tin Can and Apapa Ports Modernization Project is located within metropolitan Lagos, a rapidly growing coastal mega-city where service delivery is under sustained pressure from population growth, infrastructure constraints and environmental pollution. The Project's Area of Influence includes dense urban neighbourhoods as well as lagoon-side communities, with materially different access to basic services.

Water supply and sanitation: Across the Project setting, piped water coverage is limited and many households rely on groundwater, wells and private vendors. According to available documentation, this reliance creates ongoing concerns regarding water quality, saline intrusion in coastal areas, and contamination risk where sanitation is inadequate. Within parts of the Project Area of Influence, particularly lagoon-side and waterfront settlements, field observations indicate limited household sanitation in some locations, including reports of open defecation and discharge of untreated wastewater into adjacent water bodies. These baseline conditions present persistent public health risks and heighten sensitivity to any project-related changes in surface water quality or access to local water points.

Solid waste management and environmental cleanliness: In several communities around the ports, formal waste collection services are limited and disposal practices may be informal, including dumping in drains, along shorelines, or directly into water bodies. The wider port environment is also influenced by industrial activity, stormwater runoff from paved and yard areas, oil residues and legacy pollution. For lenders, the material interface is cumulative pressure on lagoon water quality and community health, and the requirement that the Project maintains clear control of waste and effluents within the port boundary.

Electricity and energy access: Electricity is supplied through the distribution network, but reliability is widely reported as poor. According to available documentation, some communities rely on generators and certain waterfront settlements may experience prolonged lack of stable grid access, which can increase the practical impact of any temporary disruption associated with construction logistics.

Transport and access: The ports sit within a congested urban logistics environment. In lagoon-side communities, local transport may include small boats and canoes, sometimes in poor condition, reflecting constrained infrastructure and limited safety features. This context is relevant for understanding community access patterns and sensitivities where works interact with waterways or access routes.

Overall, baseline services in the Project setting are characterised by uneven coverage and existing environmental stressors. The core implication for this NTS is that the Project must maintain strict operational discipline at the port boundary, particularly for waste, runoff and hazardous materials, so that Project activities do not add to existing public health, nuisance or pollution pressures on surrounding communities.

3.3.6 Economic Characteristics

The Tin Can and Apapa Ports Modernization Project is located in Lagos, Nigeria's largest economic centre and the country's primary maritime gateway. At national level, available documentation describes a period of strong growth in the early 2000s, followed by weaker performance from the mid-2010s onwards, compounded by external shocks (including COVID-19) and macroeconomic imbalances. Inflationary pressure and declining purchasing power have increased household vulnerability, and Nigeria remains off-track on several SDG indicators. This context is relevant because economic stress reduces the capacity of households and small businesses to absorb temporary disruption in the Project Area of Influence.

At Lagos State and port-city level, economic activity is concentrated around trade, logistics, finance and services. The Apapa/Tin Can port system underpins employment and business activity well beyond the immediate port footprint, supporting freight movement, warehousing, trading networks and a large ecosystem of formal and informal operators. Adjacent districts, including Lagos Island, function as commercial and institutional hubs with dense clusters of markets, offices, banks and services that are closely linked to port-related value chains.

Within the Project's immediate Area of Influence, the economic profile is mixed and highly interface dependent. According to available documentation, many lagoon-side communities and near-port settlements rely on informal livelihoods, including small-scale fishing and associated trading/processing activities that depend on water quality and safe access to fishing grounds; household-based artisanal activities, including boat-related crafts and small manufacturing; and petty trade and services, including transport services and micro-enterprises that depend on daily mobility, footfall and predictable access routes by road and water.

Overall, the economic baseline indicates that livelihood sensitivity in parts of the Project Area of Influence is high, with limited buffers to absorb short-term disturbance. For this NTS, the material point is that any incremental impacts would arise primarily through practical interface pathways, access constraints, traffic and safety conditions, and local pollution episodes, rather than through macroeconomic changes, and these pathways are addressed in detail in the ESIA.

Figure 13: Artisan fishing activities next to the General Support Yard

3.3.7 The Health System

Health service availability and access differ materially across the Tin Can and Apapa Ports Modernization Project area of influence. Communities closest to port operations and lagoon-side settlements face higher baseline public health risks linked to water quality, sanitation constraints and vector-borne disease, while access to primary care and emergency response is uneven. This baseline matters because project-related traffic, noise, air emissions and water pollution, if not controlled, could amplify existing vulnerabilities for sensitive receptors.

On the Lagos Island–Apapa side, healthcare infrastructure is comparatively more developed. According to available documentation, Lagos Island hosts a mix of public and private facilities, including general hospitals, maternity services, diagnostic capacity and infectious disease services. Primary healthcare centres provide basic preventive and maternal/child health services, and emergency response is supported through state ambulance services alongside private providers. Coverage and quality nonetheless vary in dense urban settings due to capacity constraints, overcrowding and workforce gaps.

In contrast, for lagoon-side communities associated with the Tin Can area of influence, available documentation indicates limited local health infrastructure, with residents often travelling to neighbouring areas for care. Where services exist, they are generally basic and, in some locations, households rely on informal providers such as traditional birth attendants. Reported barriers include transport constraints (especially during emergencies), shortages of qualified personnel and inconsistent availability of medicines and equipment. Preventive public health outreach and health education are described as limited, and the reported disease profile is consistent with local environmental conditions (notably malaria and other common infections).

For this NTS, the key point is not an inventory of facilities, but the recognition that baseline health vulnerability is elevated in some receptor communities and that impacts, if they occur, could be disproportionate. The ESIA provides the detailed baseline and supporting evidence, including where survey findings and reported disease indicators are within referenced thresholds where applicable.

3.3.8 The Education System

The Tin Can and Apapa Ports Modernization Project sits within metropolitan Lagos, but education access and day-to-day logistics are not uniform across the Project's area of influence. Conditions differ between established urban neighbourhoods linked to Lagos Island and more infrastructure-constrained port-adjacent settlements along the lagoon and waterfront. This matters because education access and attainment influence vulnerability, livelihood options, and how effectively safety and engagement messages are understood.

For the Apapa side of the Project's area of influence, available documentation describes a diversified education landscape typical of central Lagos, with a mix of public and private primary and secondary schools, alongside technical/vocational training and tertiary institutions serving the wider city. The same documentation notes common urban constraints, capacity pressure, infrastructure gaps, funding limitations, teacher shortages, and uneven access to quality, rather than an absence of education services.

For the Tin Can-related lagoon and waterfront communities within the Project's area of influence, education follows Nigeria's standard structure under the Lagos State Ministry of Education, but access is described as more constrained in several settlements. According to available documentation, public schools are limited in some communities and, where facilities are not locally present, students commute to mainland areas, including by boat in waterfront locations. This introduces practical barriers linked to cost, reliability and safety, and can affect attendance and progression. Private schooling is reported as available in some areas but affordable only to a subset of households. At state level, the documentation indicates secondary education as the most common highest attainment, and the port host communities are described as showing a similar pattern.

For this NTS, the key message is the variability: education services are more accessible in central urban areas, while lagoon-side communities face practical constraints that increase vulnerability and sensitivity to disruption. The ESIA provides the detailed baseline and supporting evidence, including survey findings and how reported education indicators compare to referenced benchmarks where applicable.

3.3.9 Cultural Heritage

The Tin Can and Apapa Ports Modernization Project is located within metropolitan Lagos, where cultural heritage is present both as physical assets and as living practices. Within the wider area of influence, particularly around Lagos Island, available documentation describes a long-established urban and trading history reflected in colonial-era buildings and streetscapes, major religious sites, and periodic cultural events and festivals. In this setting, the most plausible project interaction is indirect, through temporary changes in access, traffic patterns, and nuisance (noise and vibration), rather than direct impacts on designated heritage assets.

For the Tin Can-related waterfront and lagoon-side communities within the Project's area of influence, available documentation indicates that consultations completed to date have not identified concerns regarding impacts on cultural artifacts. The documentation also anticipates continued engagement with traditional leadership to confirm whether any culturally important places, practices, or community assets could be affected by construction activities and associated logistics, and to agree practical measures where issues are identified. For the purposes of an NTS, this should be read as an ongoing verification step based on current information, not a definitive "no impact" conclusion.

Overall, the baseline conclusion is that cultural heritage in the Project setting is primarily linked to the historic urban fabric and to community-embedded religious and cultural practices. The ESIA provides the detailed inventory, consultation record, and the procedures to manage any chance finds or community concerns that may arise during implementation.

3.3.10 Characteristics of the Affected Population

The Tin Can and Apapa Ports Modernization Project is implemented within an existing, highly industrialised port environment. The most directly exposed groups are those whose livelihoods and daily activities depend on port operations and on access to the port estate. The affected population therefore includes terminal and port-related workers and contractors operating within work areas and logistics corridors, as well as nearby users of the port-adjacent economy, including artisanal fishers, informal traders, and local transport providers using waterfront and access routes.

According to available documentation, the ESIA identifies "sensitive receptors" within the Project area of influence and confirms that exposure is primarily localised and activity-specific, depending on proximity to work sites and on reliance on the port for income. For Tin Can, the documentation notes that some port and terminal labourers, particularly casual workers, may experience temporary disruption linked to access constraints during works. A targeted socio-economic survey was undertaken in the nearest communities to support the baseline and inform the impact assessment; detailed analytical outputs are presented in the ESIA annexes.

For Apapa, the setting is similarly characterised by dense port activity within a built-up urban context. The relevant affected population remains primarily port-dependent workers and contractors, together with local waterfront users and small businesses that may experience temporary access constraints or disruption during works. Wider urban populations may experience indirect nuisance impacts (notably traffic disruption, noise and dust), but the baseline does not indicate material livelihood displacement at the city scale.

3.3.11 Traffic

Traffic congestion on the approaches to Tin Can and Apapa ports is a longstanding structural issue in Lagos, driven primarily by heavy truck volumes serving the ports and adjacent tank farms, constrained road capacity, roadside queuing/parking, and periodic roadworks. The resulting conditions include extended travel times, unreliable access for workers and local communities, higher emissions from idling vehicles, elevated noise levels, and increased road safety risks. This baseline is material from a Community Health and Safety perspective, including where congestion constrains emergency access and increases exposure of pedestrians and public transport users.

According to available documentation, the ESIA traffic assessment undertaken in March 2024 combined traffic counts (7:00–19:00 across weekdays and weekends) with interviews of road users. The findings indicate that congestion is most acute near port gates and along key corridors linking the ports to the expressway network. Respondents consistently attributed congestion primarily to port-related trucking and gate processes, with additional contributors including tank farm traffic, construction activities, driver behaviour, and weak traffic management and enforcement. The ESIA's quantitative analysis confirms that port-related truck movements are the dominant driver of congestion and associated nuisance in the port access corridors.

In this context, the key issue for the Tin Can and Apapa Ports Modernization Project is incremental impact as modernisation works will occur within an already congested transport environment. During construction, there is potential for temporary worsening of conditions if construction logistics are not tightly controlled, including movements of materials and equipment, temporary lane occupation, and increased interaction between heavy vehicles and pedestrians. During operations, the Project is not presented as a transport solution for Lagos; however, improved gate processes and yard logistics can reduce queuing and idling at the port interface, which is where the Project can credibly contribute to localised improvements. Any future change that would materially increase truck volumes would require management through operational controls and, where relevant, additional assessment.

Traffic risks will be managed through Traffic and Transport Management measures embedded in the Construction ESMP and carried into the Operational ESMP. In practical terms, this includes defined routes and timing for construction traffic, coordination with the Nigerian Ports Authority, terminal operators and relevant traffic authorities, and measures to keep port access roads passable, including for emergency services. Dust and emissions associated with traffic will be managed through basic, enforceable controls such as speed limits near receptors, vehicle maintenance and roadworthiness requirements, and housekeeping along work areas and haul routes. Performance will be tracked using indicators understandable to non-technical readers, including gate queue length and waiting time, traffic incidents, emergency access interruptions, and grievance trends.

4. IMPACT ASSESSMENT

4.1 Overview of the Method and Scope

This section summarises how the ESIA identified and evaluated the Project's potential environmental and social risks and impacts, and how mitigation and management measures were defined to avoid, minimise or reduce adverse effects.

The ESIA impact assessment focused on activities associated with site development and construction, reflecting the Project design whereby, once quay rehabilitation works are completed (including apron slab casting and installation of berth facilities), the upgraded berths are handed over to the Nigerian Ports Authority (NPA). Under this arrangement, the contractor's responsibilities end at handover, and NPA assumes responsibility for the operational management of the rehabilitated berths.

In line with good international practice, the ESIA assessed impacts by considering the interaction between Project activities and environmental and social receptors, and by characterising impacts in terms of their nature, extent, duration, frequency and reversibility, as well as the sensitivity of affected receptors. Where relevant, the ESIA distinguishes between impacts arising from planned construction activities and those associated with unplanned events (such as accidental spills) and defines mitigation and management measures intended to reduce residual impacts to an acceptable level.

Decommissioning was not assessed in detail because, according to available documentation, the infrastructure has a design life of more than 50 years and there is currently insufficient information to determine the future decommissioning approach or associated impacts with adequate precision. The ESIA therefore notes that a project-specific decommissioning assessment will need to be undertaken closer to end-of-life, to inform a Decommissioning Plan that reflects the regulatory context and good practice applicable at that time.

For this reason, the ESMP developed under the ESIA primarily addresses site development and construction, while recognising that decommissioning will require a separate, future assessment and management planning process led by the responsible port authority at the appropriate time.

Impact Assessment Overview

No	Aspect	Impact	IMPACT							Residual Impact after mitigation measures
			Duration	Extent	Frequency	Magnitude	Likelihood	Consequence	Significance	
PRE-CONSTRUCTION AND CONSTRUCTION PHASE										
1	Soil, Surface and Water Quality	Potential to impact soil and sediment quality through soil compaction, erosion, fuel and lubricant spills, improper waste disposal, and contamination from hazardous materials, which can harm nearby ecosystems and human health	Short-term to Long-term	Local	One-time to Intermittent	Small	Probable	Moderate	Moderate	Not-Significant
2		Potential Impact on coastal morphology, affecting shoreline shape, sediment dynamics, erosion patterns, beach profiles, intertidal zones, and marine habitats, with potential impacts on sediment distribution, seabed stability, and soil salinization	Short-term to Long-term	Local	One-time to Intermittent	Small	Probable	Moderate	Moderate	Minor
3		Flooding, Erosion and Sedimentation Impact. Land reclamation may increase the risk of flooding, erosion, and sedimentation by altering the natural coastline, disrupting sediment transport patterns, reducing coastal habitats, and disturbing the balance of sediment deposition and erosion.	Short-term	Local	One-time to Intermittent	Small	Probable	Minor	Minor	Non-Significant
4		impact the Marine Environment, Hydrology, and Ecosystems through habitat destruction, introduction of pollutants, alteration of water flow patterns, sediment accumulation, and potential loss of protective coastal features, affecting marine life and ecosystem services.	Short-term to Long-term	Local	One-time to Intermittent	Small	Probable	Moderate	Moderate	Minor
5		Mobilisation of Pollutants from Contaminated Land Encountered during Earthworks. When contaminated soil is disturbed, it can lead to an increase in the release of pollutants, which may include harmful substances such as heavy metals and hydrocarbons.	Short-term to Long-term	Local	One-time to Intermittent	Small	Somewhat Likely	Moderate	Minor	Non-Significant
6	Air Quality	Potential Impact ambient air quality through construction activities like demolition, concrete pouring, and heavy equipment operation, as well as transportation of materials by trucks.	Short-term	Local	One-time to Intermittent	Small	Probable	Moderate	Minor	Minor
7		The project may lead to increased dust emissions from various construction activities and ancillary sites, potentially affecting air quality	Short-term to Long-term	Local	One-time to Intermittent	Small	Probable	Moderate	Minor	Minor
8		Increase in air pollution through elevated vehicle emissions, including nitrogen oxides, volatile organic compounds, and particulate matter, which can adversely affect human health	Short-term to Long-term	Local	One-time to Intermittent	Small	Planned	Minor	Minor	Minor

No	Aspect	Impact	IMPACT							Residual Impact after mitigation measures
			Duration	Extent	Frequency	Magnitude	Likelihood	Consequence	Significance	
9		Combustion activities, including operation of diesel power generators and equipment, are key sources of emissions that can adversely affect human health through pollutants like NO2, SO2, PM10, and PM2.5.	Short-term to Long-term	Local	Intermittent	Small	Planned	Minor	Minor	Not significant
10	Noise and Vibrations	Impact on ambient noise quality from port construction activities and ancillary sites.	Short-term	Local	Intermittent	Small	Probable	Moderate	Minor	Minor
11	Noise and Vibrations	Potential Impact on Human Health Associated with Increased Noise Levels from Construction Activities to other terminal operators and contractors in the area	Short-term	Local	Intermittent	Small	Probable	Minor	Minor	Minor
12	Noise and Vibrations	Generate underwater noise and vibrations from activities such as pile driving, and vessel transportation, potentially impacting marine life and ecosystems.	Short-term	Local	Intermittent	Medium	Probable	Moderate	Moderate	Minor
13	Noise and Vibrations	Impact nearby community structures due to ground vibrations from construction activities, potentially leading to discomfort, stress, and safety risks, particularly for sensitive structures	Short-term	Local	Intermittent	Medium	Probable	Major	Major	Moderate
14	Waste	Construction activities will generate various types of waste, including construction debris, excavated soil, packaging waste, chemical waste, metal scraps, wood waste, plastic waste, electrical and electronic waste, and hazardous waste, which could potentially impact the local environment and community receptors.	Long-term	Local	Intermittent	Medium	Probable	Moderate	Moderate	Minor
15	Hydrology	Spills or Leaks Resulting in Contamination of Groundwater and Surface Water. The fuels or hydraulic fluids used in the equipment have the potential to spill, which can lead to the release of hydrocarbons. Contamination Risk: The soil, groundwater, and surface water in the area can become contaminated by these.	Short-term to Long-term	Local	One-time to Intermittent	Small	Probable	Moderate	Moderate	Minor
16	Hydrology	Activities such as quay demolition, drilling and piling, grouting, backfilling, and raw material storage and handling can lead to increased turbidity in nearby water bodies due to the resuspension of bottom sediment, production of fine particulate matter, and inadequate sediment management methods.	Long-Term	Local	Intermittent	Small	Probable	Moderate	Moderate	Minor
17	Biodiversity	Construction activities may disrupt ecosystem services such as soil stability, nutrient cycling, and habitat provision, while sediment runoff and chemical pollutants can degrade water quality and dust and emissions can impact air quality.	Long-Term	Local	One-time	Small	Unlikely	Minor	Minor	Minor

No	Aspect	Impact	IMPACT							Residual Impact after mitigation measures
			Duration	Extent	Frequency	Magnitude	Likelihood	Consequence	Significance	
18		Disturbance of Marine Ecology due to land reclamation, infrastructure development, release of sediment and construction noise and vibrations.	Long-Term	Local	One-time	Small	Unlikely	Minor	Minor	Minor
20		Introduction of Marine Invasive Species by Hull Fouling, generating ecological alterations and affecting the ecosystem services.	Long-Term	Local	One-time	Small	Unlikely	Minor	Not Significant	Not Significant
21	Social	Road Traffic and Transport Risk of Project Vehicle Traffic Accidents due to increased traffic volume during construction on employees, nearby residents, and businesses	Short-term	Local	Intermittent	Medium	Somewhat Likely	Major	Moderate	Moderate
22		Occupational health and safety issues arose construction activities including physical hazards, chemical hazards, confined spaces, exposure to dust and noise.	Long-term	Local	One-time	Small	Unlikely	Moderate	Moderate	Minor
23		Community health and safety may be impacted by physical injury risks, exposure to hazardous materials, respiratory effects from air emissions, spread of communicable diseases, changes to diet and subsistence foods, and strain on local health services and infrastructure.	Long-term	Local	One-time	Small	Unlikely	Moderate	Minor	Minor
24		Disruption on fishing activities by damaging fishing grounds, polluting waters, and altering fish habitats, leading to potential declines in fish populations and reduced income for local fishermen.	Short-term	Local	Intermittent	Medium	Probable	Major	Moderate	Minor
25		Potential Impacts on Socioeconomic opportunity, job generation, stimulation of economic growth and infrastructure improvement.	Long-term	Regional	Intermittent		Planned	N/A	Positive	Positive
26		Potential Infringement on Workers' Human Rights if not working under safe and fair working conditions	Long-Term	Regional	Intermittent	Medium	Somewhat Likely	Major	Moderate	Moderate
27		Local procurement serves as a direct investment in the immediate economy, benefiting local trading and service providers and contributing to profitability and income growth	Short-Term	Regional	Intermittent	Small	Planned	N/A	Positive	Positive
28		The influx of workers can result in competition for resources, pressure on existing infrastructure (such as healthcare, sanitation, and housing), and potential social conflicts between locals and non-locals. It will also generate job creation for local communities and an increase in local businesses' economic activity.	Medium-term	Local	Intermittent	Medium	Probable	Moderate	Moderate	Minor
OPERATION PHASE										
1	Soil, Surface and Water Quality	Cargo handling activities at the port can result in accidental spills of oils, chemicals, or other hazardous materials, which may contaminate the soil and nearby water bodies. These contaminants	Long-term	Local	Intermittent	Medium	Probable	Moderate	Moderate	Minor

No	Aspect	Impact	IMPACT							Residual Impact after mitigation measures
			Duration	Extent	Frequency	Magnitude	Likelihood	Consequence	Significance	
		can harm aquatic ecosystems and reduce water quality, making the water unsuitable for consumption or recreational use.								
2		Stormwater runoff from port operations can carry sediments and pollutants into adjacent water bodies, leading to increased turbidity and sedimentation, which may impact aquatic ecosystems and water clarity.	Long-term	Local	Intermittent	Medium	Somewhat likely	Moderate	Moderate	Minor
3	Air Quality	Continuous operation of diesel-powered equipment and vessels releases nitrogen oxides (NOx), sulphur dioxide (SO2), and particulate matter (PM), impacting air quality.	Long-term	Regional	Continuous	Medium	Probable	Moderate	Moderate	Minor
4		Handling bulk cargo, such as coal or cement, can generate dust, affecting air quality and causing respiratory issues for workers and residents.	Long-term	Local	Intermittent	Small	Somewhat likely	Minor	Minor	Not significant
5	Noise and Vibrations	Loading and unloading of cargo, as well as vessel operations, generate continuous noise that may disturb nearby residents and workers.	Long-term	Local	Continuous	Medium	Probable	Moderate	Moderate	Moderate
6		Heavy equipment movements generate vibrations that could affect nearby structures and infrastructure.	Medium-term	Local	Intermittent	Small	Unlikely	Minor	Minor	Not significant
7	Traffic	Higher vehicle movements, including trucks and workers, may cause congestion on access roads.	Medium-term	Local	Intermittent	Small	Unlikely	Minor	Minor	Not significant
8		Increased congestion raises the risk of traffic accidents and delays, impacting businesses and commuters.	Long-term	Local	Intermittent	Small	Somewhat likely	Minor	Minor	Not significant
9	Biodiversity	The operational phase will involve increased ship traffic and port operations, which may: • Introduce contaminants such as ballast water discharge, posing a risk of marine invasive species. • Disturb sediment dynamics, potentially affecting marine habitats. • Create underwater noise pollution that can disturb marine species' behaviour, particularly during vessel movements. • Impact mangrove ecosystems, which provide critical breeding grounds for fish and other marine organisms.	Long-term	Regional	Recurrent	Medium	Likely	Moderate	Moderate	Minor
10	Social	During operations, the project may introduce risks to community health and safety, including: • Increased traffic in and around the port, leading to potential road accidents. • Noise pollution from operational activities and vessel movements.	Long-term	Local	Frequent	Medium	Somewhat likely	Major	Moderate	Minor

No	Aspect	Impact	IMPACT							Residual Impact after mitigation measures
			Duration	Extent	Frequency	Magnitude	Likelihood	Consequence	Significance	
		• Air emissions from ships and port machinery, contributing to local air pollution. • Exposure to hazardous materials during cargo handling and storage. • Close interactions among workers may lead to the spread of diseases like COVID-19.								
11		Workers may face risks such as respiratory issues from emissions and noise-induced hearing loss. Workers must have access to safe working environments to protect their health and well-being.	Long-term	Local	Frequent	Medium	Somewhat likely	Major	Moderate	Minor
12		Workers may face challenges related to fair wages, overtime, and working conditions.	Long-term	Regional	Recurrent	Medium	Likely	Moderate	Moderate	Minor
13		The operational phase will create significant employment opportunities and stimulate the local economy. Positive impacts include: • Direct employment in port operations, cargo handling, and logistics. • Indirect employment through support services such as transportation, food vendors, and retail businesses. • Economic growth from increased trade activities, benefiting the local community and businesses.	Long-term	Regional	Intermittent	Large	Planned	N/A	Positive	Positive

Mitigation measures for each impact have been proposed and are considered for the development of the project ESMP and related Management Plans.

5. ENVIRONMENTAL AND SOCIAL MANAGEMENT PLAN

The ESIA presents an outline Environmental and Social Management Plan (ESMP) that summarises the avoidance, minimisation and mitigation measures required to manage the potential environmental and social impacts of the Tin Can and Apapa Ports Modernization Project. The ESMP consolidates the measures identified in the Impact Assessment and is intended to support consistent implementation of controls across the Project footprint and associated ancillary areas during site development and construction works.

The ESMP's overall objective is to provide a practical tool that clearly summarises and lists the mitigation and management measures committed to in the ESIA. It therefore serves as the primary reference document linking identified risks and impacts to the specific controls, monitoring actions, and management procedures to be applied on site, including measures relevant to unplanned events such as spills and other incidents.

The ESMP will be implemented by the construction joint venture and its subcontractors as part of day-to-day site management, under the oversight arrangements set out for the Project. Following completion and handover of the rehabilitated quay infrastructure, the Nigerian Ports Authority (NPA) will assume responsibility for the operational phase of the port facilities.

The ESMP is supported by topic-specific management plans and procedures that address the key environmental and social aspects relevant to quay rehabilitation in a highly industrialised, urban coastal setting. These plans cover, as applicable, areas such as dust and air emissions control, environmental construction methods, waste and wastewater management, contaminated land management, soil, runoff and sediment control, occupational health and safety, communicable disease management, community health and safety, contractor and supplier management, stakeholder engagement and grievance management (including dedicated community and worker channels), traffic and transport management, security management, and interface/coordination arrangements with NPA, terminal operators and other port users.

Environmental and Social Management Plan

N	Aspect	Potential Impact	Impact Magnitude	Mitigation Measures	Residual Impact
1	Soil, Surface and Water Quality	Potential to impact soil and sediment quality through soil compaction, erosion, fuel and lubricant spills, improper waste disposal, and contamination from hazardous materials, which can harm nearby ecosystems and human health	Moderate	All the construction material will be stored in dedicated storage areas; A designated machinery and equipment storage area will be developed for the Project;	Not-Significant
2				Storage of construction materials, fuels, lubricants, and hazardous materials in dedicated areas with secondary containment	
3				Removal of temporary structures, surplus materials, and waste upon completion of work Use of construction and demolition waste for site filling	
4		Potential Impact on coastal morphology, affecting shoreline shape, sediment dynamics, erosion patterns, beach profiles, intertidal zones, and marine habitats, with potential impacts on sediment distribution, seabed stability, and soil salinization	Moderate	Establish drainage structures and stormwater system to minimize erosion and maintain natural features. Stormwater shall be separated from sanitary (domestic) and industrial wastewater streams in accordance with the WWMP. Stormwater from potentially contaminated areas (e.g., workshops, fuel storage, wash bays, batching/washout areas) shall be captured and routed through silt traps and/or pre-treatment units (grease traps / oil–water separators) prior to discharge to the stormwater network. Surface runoff from process areas or potential sources of contamination should be prevented, or where not practical, runoff from process and storage areas should be segregated from potentially less contaminated runoff. Continuously monitor shoreline changes and adapt management strategies accordingly.	Minor
5				Establishment of a Stormwater system. Stormwater runoff from process/maintenance areas shall be routed through silt traps and, where relevant, grease traps/oil–water separators before entering the stormwater network. No untreated wastewater (domestic or industrial) shall be discharged to Badagry Creek / Lagos Lagoon Leverage rainwater and flood resources to establish a long-term leaching mechanism. This helps reduce soil salinity over time	
6				Analyse soil properties before disposal to ensure compatibility with the seabed. Establish Supplier Management Plan to ensure sustainable sources of material and undertake monitoring and audits to suppliers.	
7				Maintenance or repair of mechanized fleet to primarily take place in dedicated workshop facilities located at the Savaric Yard (Ibafon) and the General Support Yard within Tin Can Port, each with bunded, impervious floors, oil traps and a wash bay connected to a closed septic tank. Septic tanks will be managed in line with the Project Wastewater Management Plan and emptied periodically by a licensed, LAWMA-approved contractor; no direct discharge to soil or	

N	Aspect	Potential Impact	Impact Magnitude	Mitigation Measures	Residual Impact
				surface water will be allowed. Oil-water separators and grease traps should be installed and maintained as appropriate at refuelling facilities, workshops, wash bays, fuel storage and containment areas.	
8		Flooding, Erosion and Sedimentation Impact. Land reclamation may increase the risk of flooding, erosion, and sedimentation by altering the natural coastline, disrupting sediment transport patterns, reducing coastal habitats, and disturbing the balance of sediment deposition and erosion.	Minor	Erosion Control: Sediment control measures are instead focused on localized, controlled in-water activities Sediment Management: Seabed disturbance will be minimal, targeted, and performed using a barge-mounted long-reach excavator under controlled conditions. Diver supervision and real-time positioning (RTK/DGPS) will ensure precise placement and minimize turbidity. If small quantities of sediment are displaced during mattress installation, they will be contained within the immediate work area. Should any pockets of contaminated sediment be encountered, these will be segregated, placed in lined skips or barges, and transported by licensed contractors to LASEPA-approved facilities. Continuous monitoring will ensure no release of contaminated material into the lagoon. Flood Management: Develop flood management plans and install drainage systems.	Non-Significant
9		impact the Marine Environment, Hydrology, and Ecosystems through habitat destruction, introduction of pollutants, alteration of water flow patterns, sediment accumulation, and potential loss of protective coastal features, affecting marine life and ecosystem services.	Moderate	Habitat Protection: Identify and protect critical marine habitats. Pollution Prevention: Implement spill prevention plans and use eco-friendly materials, controlled placement techniques, spill-prevention measures, and water-quality monitoring Flow Management: Maintain natural water flow patterns during construction. Pollution prevention and sediment controls are governed by the CESMP and SWMP; no separate BMP applies during construction. Wastewater controls (domestic septic tanks; industrial pre-treatment via grease traps/oil–water separators/sedimentation pits; stormwater segregation) shall be implemented in accordance with the WWMP to prevent contaminated discharges to the lagoon	Minor
10		Mobilisation of pollutants from contaminated land or sediments can occur if earthworks disturb legacy contamination. Baseline studies of the Lagos Lagoon and Tin Can Island area indicate the presence of heavy metals (e.g., Cu, Pb, Zn, Cd) and hydrocarbons in sediments due to long-term industrial activity. Only very limited excavation will take place in this Project, restricted to the	Minor	Soil Testing: Conduct pre-construction soil tests to identify contamination. Containment Measures: Use liners and barriers to prevent pollutant spread. Safe Handling Procedures: Implement procedures for safe handling and disposal of contaminated soil. Protective Gear: Ensure workers use appropriate protective gear.	Minor

N	Aspect	Potential Impact	Impact Magnitude	Mitigation Measures	Residual Impact
		controlled removal of damaged quay wall elements, localized seabed preparation for scour protection, and minor excavation inside the existing construction yards. No bulk earthworks or land reshaping will occur. All works take place on existing paved port platforms.			
11	**Air Quality**	Potential Impact ambient air quality through construction activities like demolition, concrete pouring, and heavy equipment operation, as well as transportation of materials by trucks.	Minor	Dust Control: Implement water spraying and other dust suppression techniques. Covering Materials: Cover materials during transport to minimize dust release. Regular Maintenance: Maintain equipment to ensure optimal performance and reduced emissions. Emissions Control: Use low-emission vehicles and machinery.	Minor
12		The project may lead to increased dust emissions from various construction activities and ancillary sites, potentially affecting air quality	Minor	Dust Suppression: Implement water spraying and other dust control measures. Covering Materials: Cover stockpiles and trucks to minimize dust. Paved Roads: Use paved or treated roads to reduce dust from vehicle movement.	Minor
13		Increase in air pollution through elevated vehicle emissions, including nitrogen oxides, volatile organic compounds, and particulate matter, which can adversely affect human health	Minor	Emission Standards: Enforce strict emission standards for vehicles. Alternative Fuels: Promote the use of alternative fuel vehicles. Vehicle Maintenance: Regularly maintain vehicles to ensure optimal performance and lower emissions.	Minor
14		combustion activities, including operation of diesel power generators and equipment, are key sources of emissions that can adversely affect human health through pollutants like NO2, SO2, PM10, and PM2.5.	Minor	Emission Control: Use emission control devices like diesel particulate filters and catalytic converters. Low-Emission Equipment: Utilize low-emission or alternative fuel equipment, if possible. Regular Maintenance: Conduct regular maintenance of generators and equipment to ensure optimal performance. Dust Control: Implement dust suppression measures.	Not significant
15	**Noise and Vibrations**	Impact on ambient noise quality from port construction activities and ancillary sites.	Minor	Noise Barriers: Install temporary noise barriers around construction zones. Low-Noise Machinery: Utilize equipment with noise reduction features. Work Scheduling: Schedule noisy activities during off-peak hours. Monitoring and Communication: Regularly monitor noise levels and inform the community about construction schedules.	Minor

N	Aspect	Potential Impact	Impact Magnitude	Mitigation Measures	Residual Impact
				Work Scheduling: Noisy activities (e.g., pile driving, demolition, heavy equipment use) will be scheduled during off-peak hours to minimise disruption to both workers and nearby communities. For communities, this reduces exposure to construction noise during sensitive times such as evenings, nights, and weekends. Advance notice will be provided to affected residents and businesses, and noise monitoring will be undertaken at the site boundary to ensure compliance with permissible limits.	
16		Potential Impact on Human Health Associated with Increased Noise Levels from Construction Activities to other terminal operators and contractors in the area	Minor	Noise Barriers: Install temporary noise barriers around construction sites. Quiet Equipment: Use machinery with noise reduction features. Work Scheduling: Schedule noisy activities during times that least impact other operators and contractors. Communication: Inform nearby operators and contractors about the construction schedule.	Minor
17		Generate underwater noise and vibrations from activities such as pile driving, and vessel transportation, potentially impacting marine life and ecosystems.	Moderate	Noise Abatement Technologies: Use noise dampening techniques and equipment like bubble curtains around pile driving sites. Scheduling: Conduct noisy activities during times least sensitive to marine life. Speed Restrictions: Implement speed limits for vessels to minimize underwater noise.	Minor
18		Impact nearby community structures due to ground vibrations from construction activities, potentially leading to discomfort, stress, and safety risks, particularly for sensitive structures. Construction activities such as pile driving, compaction, and demolition may generate ground vibrations that could affect nearby community structures and cause discomfort or stress to residents. Sensitive structures (e.g., informal housing, clinics, schools, or older buildings with compromised foundations) may be particularly vulnerable.	Major	Scientific Analysis: Prior to commencement, a vibration risk assessment will be conducted, including: Pre-construction structural surveys of buildings within the zone of influence; Establishment of vibration thresholds based on international standards (e.g., DIN 4150, BS 7385) and IFC EHS Guidelines; Modelling of expected vibration levels from planned equipment and methods. Vibration Control: Implement vibration monitoring and control techniques. Sensitive Structure Surveys: Conduct pre-construction surveys of nearby sensitive structures. Construction Scheduling: Schedule construction activities to minimize impact during sensitive times. Community Communication: Inform the community about construction activities and schedules.	Moderate
19	**Waste**	Construction activities will generate various types of waste, including construction debris, excavated soil,	Moderate	Waste Management Plan: Develop and implement a comprehensive waste management plan. Segregation and Recycling: Segregate waste streams and maximize recycling. Proper Disposal: Ensure proper disposal of hazardous and non-hazardous waste.	Minor

N	Aspect	Potential Impact	Impact Magnitude	Mitigation Measures	Residual Impact
		packaging waste, chemical waste, metal scraps, wood waste, plastic waste, electrical and electronic waste, and hazardous waste, which could potentially impact the local environment and community receptors		Training Programs: Train workers on waste management practices.	
20	**Hydrology**	Spills or Leaks Resulting in Contamination of Groundwater and Surface Water. The fuels or hydraulic fluids used in the equipment have the potential to spill, which can lead to the release of hydrocarbons. Contamination Risk: The soil, groundwater, and surface water in the area can become contaminated by these	Moderate	Spill Prevention Plans: Develop and implement spill prevention and response plans. Secondary Containment: Use secondary containment systems for fuel and hydraulic fluid storage. Regular Maintenance: Regularly inspect and maintain equipment to prevent leaks. Emergency Response Training: Train staff in spill response and containment procedures.	Minor
21		Activities such as quay demolition, drilling and piling, grouting, backfilling, and raw material storage and handling can lead to increased turbidity in nearby water bodies due to the resuspension of bottom sediment, production of fine particulate matter, and inadequate sediment management methods.	Moderate	Silt curtains will be used only where safe and operationally feasible, considering port currents, vessel movements, navigation lanes, and berthing operations. If conditions do not allow deployment, equivalent sediment control measures will be applied — including precise mechanical excavation, timing during low current periods, and turbidity monitoring. Dust Suppression: Use water spraying and dust control methods. Sediment Management: Seabed disturbance will be minimal, targeted, and performed using a barge-mounted long-reach excavator under controlled conditions. Diver supervision and real-time positioning (RTK/DGPS) will ensure precise placement and minimize turbidity. If small quantities of sediment are displaced during mattress installation, they will be contained within the immediate work area. Should any pockets of contaminated sediment be encountered, these will be segregated, placed in lined skips or barges, and transported by licensed contractors to LASEPA-approved facilities. Continuous monitoring will ensure no release of contaminated material into the lagoon. Material Handling Protocols: Ensure proper storage and handling of raw materials. All concrete washout water, drilling/casing slurry water (if any), and wash water shall be managed via WWMP controls (sedimentation/decantation pits + controlled disposal) to prevent turbidity and contaminated runoff	Minor
22	**Biodiversity**	construction activities may disrupt ecosystem services such as soil stability, nutrient cycling, and habitat provision, while sediment runoff and chemical	Minor	Nutrient Management: Preserve and enhance nutrient cycling through soil conservation practices. Habitat Protection: Identify and protect key habitats. Pollution Prevention: Proper handling and disposal of chemical pollutants. Dust Suppression: Water spraying and dust control methods.	Minor

N	Aspect	Potential Impact	Impact Magnitude	Mitigation Measures	Residual Impact
		pollutants can degrade water quality and dust and emissions can impact air quality		Emission Control: Use low-emission equipment and vehicles.	
23		Disturbance of Marine Ecology due to land reclamation, infrastructure development, release of sediment and construction noise and vibrations.	Minor	• Implement quiet construction methods and ensure equipment is well maintained to minimize underwater and ambient noise and vibration. • Schedule works to avoid unnecessary disturbance during peak tidal or high-current conditions where practicable. • Prevent uncontrolled release of sediments, debris, cement residues, or pollutants from quay-side works through good housekeeping, spill prevention, and containment measures. • Implement sediment and runoff controls in accordance with the Soil and Water Management Plan (SWMP), including drainage management, interception of runoff, and prevention of direct discharge to marine waters. • Wastewater and wash water from quay-side activities shall be managed in accordance with the WWMP (segregation of streams; pre-treatment where relevant; no untreated discharge). No in-water construction, dredging, or land reclamation activities are planned; therefore, no standalone Biodiversity Management Plan applies during the construction phase.	Minor
24	Social	Road Traffic and Transport Risk of Project Vehicle Traffic Accidents due to increased traffic volume during construction on employees, nearby residents, and businesses	Moderate	Traffic Management Plan: Implement traffic control measures, designated routes, and schedules. Driver Training: Ensure all drivers are trained in safe driving practices. Vehicle Maintenance: Regular maintenance checks for all project vehicles. Community Awareness: Inform nearby residents and businesses about traffic changes.	Moderate
25		Occupational health and safety issues arose construction activities including physical hazards, chemical hazards, confined spaces, exposure to dust and noise.	Moderate	Implement Safety Protocols: Regular training, PPE, and safety gear. Chemical Handling Procedures: Proper storage and use of chemicals. Confined Spaces Protocol: Ensure safe entry and exit procedures. Dust and Noise Control: Dust suppression and noise barriers.	Minor
26		Community health and safety may be impacted by physical injury risks, exposure to hazardous materials, respiratory effects from air emissions, spread of communicable diseases, changes to diet and subsistence foods,	Minor	Safety Protocols: Strict adherence to prevent injuries. Hazardous Material Management: Proper handling and disposal. Air Quality Control: Dust suppression and emission reduction. Health Programs: Screenings and vaccinations. Diet Support: Ensure nutritious food availability. Infrastructure Upgrade: Improve local health services.	Minor

N	Aspect	Potential Impact	Impact Magnitude	Mitigation Measures	Residual Impact
		and strain on local health services and infrastructure.			
27		Disruption on fishing activities by damaging fishing grounds, polluting waters, and altering fish habitats, leading to potential declines in fish populations and reduced income for local fishermen.	Moderate	Environmental Monitoring: Regular water quality checks and assessments of fish populations, controlled placement techniques, spill-prevention measures, and water-quality monitoring Engage with Local Fishermen: Develop compensation plans and alternative livelihood programs.	Minor
28		Potential Impacts on Socioeconomic opportunity, job generation, stimulation of economic growth and infrastructure improvement.	Positive	-	Positive
29		Potential Infringement on Workers' Human Rights if not working under safe and fair working conditions	Moderate	Implement a Robust Health and Safety Program: Regular training, provision of safety gear, and strict adherence to safety protocols. Fair Working Conditions: Ensure compliance with labour laws, provide fair wages, and establish clear grievance mechanisms. Regular Audits and Inspections: Conduct frequent audits and inspections to ensure compliance with safety and labour standards.	Moderate
30		Local procurement serves as a direct investment in the immediate economy, benefiting local trading and service providers and contributing to profitability and income growth	Positive	-	Positive
31		The influx of workers can result in competition for resources, pressure on existing infrastructure (such as healthcare, sanitation, and housing), and potential social conflicts between locals and non-locals. It will also generate job creation for local communities and an increase in local businesses' economic activity.	Moderate	Develop a Local Hiring Policy: Prioritize hiring from local communities to ensure job creation benefits them directly. Improve Infrastructure: Invest in upgrading local healthcare, sanitation, and housing to accommodate increased demand. Community Engagement Programs: Organize regular meetings with local communities to address concerns and foster good relationships. Resource Management Plan: Implement measures to manage the consumption of local resources effectively. Conflict Resolution Mechanism: Establish a system to address and resolve conflicts between workers and locals.	Minor

N	Aspect	Potential Impact	Impact Magnitude	Mitigation Measures	Residual Impact
32	**Resource Efficiency**	Excessive water consumption leading to strain on local supply; inefficient energy use resulting in higher greenhouse gas emissions and costs.	Moderate	Install water-saving fixtures. Reuse non-potable water for dust suppression and equipment washing, if needed. Ensure metered supply. Enforce idling controls for vehicles and machinery.	Not significant

The following management plans are to be prepared as standalone documents. It is crucial to note that these management plans should be read and implemented together, as they are interdependent and complement each other:

Dust Management Plan: This plan identifies sources of dust, outlines measures to control emissions, and includes monitoring strategies to protect air quality and public health.
Environmental Construction Method Statement: This statement describes environmentally friendly construction methods and practices to minimize environmental impacts during project implementation.
Emergency Management Plan: This plan outlines procedures for responding to emergencies, including natural disasters and accidents, to minimize harm to people, property, and the environment.
Human Resources Management Plan: This plan outlines strategies for managing human resources, including recruitment, training, and welfare, to ensure a safe and healthy work environment.
Waste Management Plan: This procedure outlines the proper management of construction and hazardous waste, including storage, handling, transportation, and disposal, to minimize environmental impacts.
Wastewater Management Plan: Establishes controls for the collection, treatment, and disposal of wastewater generated during construction activities to prevent pollution of surface water, drainage systems, and the marine environment.
Construction Contaminated Land Management Plan: This plan outlines procedures for managing contaminated land during construction activities, including remediation measures to protect human health and the environment.
Soil and Water Management Plan: Details sediment control, runoff management, drainage protection, and pollution prevention measures to safeguard soil and surface water resources during construction. Measures relating to sediment control and protection of the marine environment referenced in the CESMP are implemented through the SWMP.
Occupational Health and Safety Management Plan: This plan outlines measures to ensure the health and safety of workers, including training, protective equipment, and monitoring, in accordance with international standards.
COVID-19 Management Procedure: This procedure outlines measures to prevent and control the spread of COVID-19 among workers and the community, including hygiene practices, tests and awareness campaigns.
Community Health and Safety Management Plan: This plan addresses health and safety concerns of local communities near the project site, focusing on mitigating risks from construction activities and promoting community well-being.
Contractor and Supplier Management Plan: This plan outlines procedures for selecting, contracting, and managing contractors and suppliers to ensure compliance with environmental, social, and health and safety requirements.
Stakeholder Engagement Plan: This plan outlines strategies for engaging with stakeholders, including local communities, to address their concerns and ensure their participation in decision-making processes. It should include a Community Grievance Mechanism.

Community and Workers Grievance Mechanism: Provides accessible and transparent channels for workers and external stakeholders to raise concerns or complaints related to construction activities, working conditions, or environmental and social impacts, and ensures timely resolution.
Traffic Management Plan: Establishes measures to manage construction traffic, including routing, signage, speed control, and coordination with port authorities, to reduce safety risks and disruption to other port users.
Security Management Plan: Defines security arrangements for the Project, including access control, coordination with public security forces where applicable, and conduct requirements for security personnel, consistent with human rights principles.
Interface and Coordination Management Plan: Describes coordination mechanisms between the Project, port operators, terminal users, and other third parties operating within the port environment to ensure safe and efficient interface management during construction.

6. CONCLUSION

The Tin Can and Apapa Ports Modernisation Project is intended to rehabilitate and upgrade existing port infrastructure to restore safe, reliable and efficient operations within an already industrialised port setting. The works are limited to the existing port estates and are designed to support national and Lagos State trade and logistics functions, with associated opportunities for employment and local procurement during implementation.

The ESIA identifies the main impact pathways as construction-related emissions and nuisance (dust, noise and vibration), soil/sediment and water quality risks linked to earthworks and marine activities, traffic and road safety pressures in a congested port-access environment, and community and workforce health and safety. Based on the ESIA's impact evaluation, most planned impacts are rated Low to Moderate before mitigation and reduce to Minor or Not Significant after the application of the defined management measures. Positive effects are reported for jobs and economic activity, including direct and indirect employment and local supply-chain demand.

A limited number of residual impacts remain Moderate after mitigation, notably:

1. construction-related ground vibration effects on nearby sensitive structures;
2. traffic and transport accident risk associated with project vehicle movements;
3. labour and working conditions risks if not effectively managed;
4. operational noise in the active port environment.

The ESIA also addresses accidental events such as spills through defined prevention and emergency response measures. Overall, the ESIA conclusion is that the Project can proceed with predominantly minor and non-significant residual effects when the specified mitigation and management measures are implemented, subject to the required regulatory approvals.